
When Shift Happens - Confounding Is to Blame

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Distribution shifts inject uncertainty that systematically erodes the robustness and
2 generalization of machine learning models. Although the prevailing view holds
3 that learning causal-invariant representations is the route to robustness under such
4 shifts, a growing body of empirical evidence reveals a striking counterpoint: (i)
5 empirical risk minimization (ERM) consistently matches, and frequently surpasses,
6 state-of-the-art out-of-distribution (OOD) generalization methods, and (ii) its OOD
7 generalization performance improves precisely when all available covariates—not
8 merely the causal ones—are exploited. We argue that a single, largely overlooked
9 mechanism—shifts in *hidden confounders*—offers a coherent and unifying expla-
10 nation for both observations. When the distribution of an unobserved common
11 cause changes across environments, it induces coupled shifts in the observed co-
12 variate and outcome distributions that jointly violate the structural assumptions
13 on which existing OOD generalization approaches rest. For this regime we prove
14 that maximizing the predictive information between a model’s outputs and the true
15 outcome reduces to learning *environment-specific* relationships, so that invariant
16 representations, while necessary, are provably insufficient on their own. We further
17 establish that augmenting a model with proxies for the hidden confounders prov-
18 ably shifts the governing information-theoretic quantities in favorable directions,
19 thereby mitigating the difficulty that confounding shift creates. Empirically, across
20 eight real-world tabular benchmarks and a suite of controlled synthetic studies, the
21 statistical signature of hidden confounding shift is pervasive, the decomposition
22 term our theory identifies as decisive is positively associated with both in- and
23 out-of-distribution accuracy, and enriching a model with informative non-causal
24 covariates improves generalization. Together, these results supply theoretical
25 foundations and actionable guidance for the design of robust OOD generalization
26 algorithms and principled covariate-selection strategies.

1 Introduction

27 Generalization—the capacity to draw reliable conclusions about unseen data from observed data—is
28 a foundational concern across the sciences, recurring under different names but a common underlying
29 logic. It appears in medicine and the social sciences as *external validity*, the property that certifies
30 the transfer of findings obtained in one population to another [13]; in ecology as *space-for-time*
31 *substitution*, in which spatial variation stands in as a proxy for temporal change to infer long-term
32 dynamics [60]; and in engineering as *robust control*, in which a system must preserve its performance
33 under unmodeled disturbances [35]. Over the past decade, machine learning has emerged as a
34 principal instrument for learning generalizable models [100, 88, 49], which lends the question of
35 *when* and *why* such models generalize both practical urgency and theoretical interest.

36 Reliable generalization requires that a model remain robust to the distribution shifts that separate
37 training from deployment. A model trained to predict *food stamp reciprocity* from *household attributes*
38 in one state, for instance, should retain its predictive performance when deployed in another. We
39 say that a model attains *out-of-distribution (OOD) generalization* when it sustains performance
40 on both in-distribution (ID) data—the distribution from which training samples are drawn—and
41 out-of-distribution (OOD) test data. To this end, several distinct families of methods—domain
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generalization [56, 31, 53, 99, 6, 38, 74], domain adaptation [98, 82, 51, 90, 91, 81], and robust learning [42, 69, 94]—furnish provable OOD guarantees under specific structural assumptions. A persistent and reproducible finding, however, cuts against this program: under careful model selection [30, 38, 87], models trained by standard *empirical risk minimization (ERM)* [86] attain competitive—and frequently superior—OOD generalization across a broad spectrum of real-world applications [30, 24, 50, 57, 67, 87]. More striking still, a recent study [57] spanning 16 real-world tabular datasets establishes that predicting the outcome from *all* available covariates—irrespective of whether they directly affect it—improves OOD generalization. Because these findings run directly counter to the prevailing intuition that discarding non-causal covariates is the safe choice, they call for a closer examination of the mechanisms that actually govern OOD generalization.

Distribution shifts arise characteristically when data are collected across heterogeneous environments [71]. *Food stamp reciprocity*, for example, varies from state to state precisely because each administers distinct eligibility rules, yielding environment-specific distributions. Within such data, a subset of the statistical relationships between covariates and outcome may persist unchanged across every environment; these are the *invariant relationships* [6], and identifying and learning them is a well-established route to OOD generalization [6, 59, 31, 66, 56, 98, 63]. A particularly strong form is *causal invariance*, in which the causal relationships themselves remain invariant across environments [71, 59, 6]. In practice, however, not all covariates relevant to predicting an outcome are observed—whether because of data-collection limits, privacy constraints, or measurement error [14, 52, 20, 48]—and their absence can sever the very invariant relationships on which generalization is predicated. The difficulty is compounded when the unobserved variables are *confounders*: variables that influence both the observed covariates and the outcome. Such hidden confounders are pervasive, and shifts in their distributions demarcate distinct environments. Left unaddressed, these shifts do more than degrade generalization [40, 4, 85, 62]; they can drive a model to learn outright incorrect relationships between the observed covariates and the outcome—an effect we illustrate concretely in § 3.

Despite its centrality, the effect of hidden confounding shift on OOD generalization has received comparatively little systematic treatment [4, 40, 85, 62]. This work addresses that gap. We develop a unified information-theoretic and causal account of hidden confounding shift and, drawing on it, offer a coherent explanation for the otherwise puzzling findings accumulating in the recent literature [30, 24, 50, 57, 67, 87]. Our contributions are as follows:

- **A causal characterization of confounding shift.** We show, from a causal perspective, why a shift in a hidden confounder simultaneously induces label, covariate, conditional-covariate, and concept shifts, and thereby why it constitutes a distinct and analytically central regime for understanding OOD generalization (§ 3); the same lens explains why augmenting a model with informative, non-causal covariates helps.
- **Environment-specific learning is provably necessary.** We prove that, under hidden confounding shift, maximizing the predictive information between model predictions and true outcomes reduces to maximizing an *environment-specific* quantity. Invariant representations, while necessary, are therefore provably insufficient on their own (§ 4.1, § 4.2).
- **Informative covariates provably help.** We prove that covariates informative of either the outcome or the hidden confounders increase the predictive information between model predictions and true outcomes, placing principled covariate selection—rather than blanket removal of non-causal features—on a firm theoretical footing under hidden confounding shift (§ 4.3).
- **Converging empirical evidence.** Across eight real-world tabular benchmarks and controlled synthetic studies, our experiments provide mutually reinforcing evidence that (i) the statistical signature of hidden confounding shift is pervasive in real-world tabular data, (ii) the degree to which a model learns environment-specific relationships is positively associated with both ID and OOD test accuracy, and (iii) incorporating informative, non-causal covariates improves generalization (§ 5).

Taken together, these results offer a principled explanation for the surprising effectiveness of ERM under real-world distribution shifts and identify hidden confounding shift as a mechanism that OOD generalization methods must confront directly.

2 Related work

Out-of-distribution (OOD) generalization spans several closely related problem settings, including domain generalization [56, 31, 53, 99, 6, 38], domain adaptation [98, 82, 51, 90, 91, 81], robust learning [42, 69, 94], federated learning [54, 36, 44, 97, 45], and OOD detection [41, 32, 92, 84, 49]. The settings differ chiefly in the information they assume at training and test time: domain adaptation presumes access to unlabeled test-set data, whereas domain generalization relies on environment labels during training; federated learning operates within a collaborative framework that must additionally reconcile communication efficiency and privacy across distributed environments; and OOD detection targets the identification of samples that deviate markedly from the training distribution. A recurring objective across many of these methods is to recover relationships that remain invariant across environments. A growing line of work, however, shows that invariance alone is insufficient and that additional inductive biases are needed for improved generalization [46, 72, 93]; Ye et al. [93], in particular, argue that feature invariance is necessary but not sufficient and foreground the complementary role of feature *informativeness*. We emphasize that these methods vary widely in how strictly they enforce invariance—from hard causal-invariance constraints to soft penalties on cross-environment risk variance or feature discrepancy—and consequently behave differently under confounding shift. Our analysis unifies invariance and informativeness within a single information-theoretic account, making explicit how the two desiderata jointly determine OOD generalization and how each is affected by hidden confounding shift.

Proxy variable adjustment: For the setting in which confounders are observed during training but not at test time, Landeiro and Culotta [40] adjust for confounding shift to improve classifier performance. Building on proxy-based adjustment for causal effect estimation [55, 39], recent domain adaptation methods recast unknown distribution shifts as a causal-effect identification problem [4, 85], and OOD generalization under hidden confounding shift has been studied under an overlapping-confounder-support assumption [62]. These approaches prescribe adjustment procedures; our contribution is complementary, in that we characterize the *mechanism by which* proxy variables enhance OOD generalization—quantifying, through the information-theoretic decomposition of § 4, when and why such adjustments succeed.

All-variable models vs. causal models. Nastl and Hardt [57] introduced a benchmark study that partitions covariates into four groups—causal (conservatively chosen), arguably causal, anti-causal, and other spurious covariates—and demonstrated that, across 16 benchmark datasets, models using all covariates Pareto-dominate those restricted to causal or arguably causal subsets on both ID and OOD data. These robust empirical results have wanted for a theoretical explanation; we provide one, identifying the scenarios and mechanisms under which they arise. For linear causal models, anchor regression [68] interpolates between two estimation paradigms—models that admit all observed covariates and models that retain only causal covariates—and our analysis clarifies the effect of admitting additional, not necessarily causal, covariates under hidden confounding shift. Eastwood et al. [21] show that unstable covariates can improve performance when they carry label information, provided they are conditionally independent of the stable covariates given the label, adjusting for the shift by inspecting the unlabeled test domain. Notably, when their method is applied to a medical real-world dataset not constructed for this problem [8], ERM remains competitive, consistent with Nastl and Hardt [57]—a pattern that aligns with the broader result that, under *well-specified* covariate shift, maximum likelihood estimation (MLE) is minimax-optimal for OOD generalization [26]. Real-world settings, however, are frequently *not* well-specified, and hidden confounding shift—the phenomenon at the center of this work—is one important reason why.

3 Manifestations of hidden confounding shift

This section develops the background on hidden confounding shift and, through an example grounded in the causal-effect identification of the observed covariates on the outcome, motivates why the phenomenon merits direct treatment.

Types of distribution shifts. For covariates $\mathbf{X}$ and target Y , two environments e and e' can differ along any of the distributional dimensions enumerated in Table 1. A shift along any of these dimensions can be traced to a common source: a change in the distribution $\mathbb{P}(U)$ of

Table 1: Distribution shifts that can be induced across two environments e and e' .

Type of Shift	Mathematical Expression
Label	$\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$
Covariate	$\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$
Conditional Covariate	$\mathbb{P}^e(\mathbf{X} Y) \neq \mathbb{P}^{e'}(\mathbf{X} Y)$
Concept	$\mathbb{P}^e(Y \mathbf{X}) \neq \mathbb{P}^{e'}(Y \mathbf{X})$
Dataset	$\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$

an unobserved covariate U that causes $\mathbf{X}$, Y , or both (see Figure 1). Such a change propagates into the observed distributions over $\mathbf{X}$ and Y through the marginalization $\mathbb{P}(\mathbf{X}) = \sum_U \mathbb{P}(U) \mathbb{P}(\mathbf{X} | U)$: whenever $\mathbb{P}^e(U) \neq \mathbb{P}^{e'}(U)$ and $U \rightarrow \mathbf{X}$, it follows that $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$. The same argument accounts for the induced shifts in $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, and $\mathbb{P}(\mathbf{X}, Y)$.

Many established OOD generalization methods presuppose a particular causal relationship among U , $\mathbf{X}$, Y that guarantees a specific invariance. When $U \rightarrow Y \rightarrow \mathbf{X}$ (Figure 1(a)), the factorization $\mathbb{P}(\mathbf{X}, Y) = \mathbb{P}(Y) \mathbb{P}(\mathbf{X} | Y)$ implies that any joint shift $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$ is attributable solely to label shift, $\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$,

while the conditional covariate distribution remains invariant, $\mathbb{P}^e(\mathbf{X} | Y) = \mathbb{P}^{e'}(\mathbf{X} | Y)$, because $\mathbb{P}(\mathbf{X} | Y, U) = \mathbb{P}(\mathbf{X} | Y)$ [89, 83, 25, 5, 47, 96, 18, 79]. Symmetrically, when $U \rightarrow \mathbf{X} \rightarrow Y$ (Figure 1(b)), the joint shift is attributable to covariate shift, $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$, while the concept distribution stays invariant, $\mathbb{P}^e(Y | \mathbf{X}) = \mathbb{P}^{e'}(Y | \mathbf{X})$ [70, 16, 80, 11, 28]. Each of these convenient invariances is what its associated family of methods exploits. Both, however, break down in the more general case in which U causes *both* $\mathbf{X}$ and Y (Figure 1(c))—a structure that recent work argues is common in practice [50, 4, 40, 85, 62, 65, 64]. Here a single shift in $\mathbb{P}(U)$ can trigger label, covariate, conditional-covariate, and concept shift *simultaneously*. The only invariance that confounding shift preserves is $\mathbb{P}^e(\mathbf{X}, Y | U) = \mathbb{P}^{e'}(\mathbf{X}, Y | U)$ —the *stable confounding shift* condition [85, 4]—and because U is unobserved, this invariance cannot be exploited directly in the observed data. It is precisely this entanglement of shifts behind an unobservable stabilizing variable that makes confounding shift qualitatively harder than the pure label- or covariate-shift settings for which existing methods were designed.

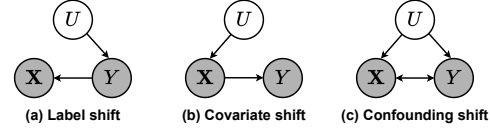


Figure 1: Causal graphs underlying distribution shifts.

Information-theoretic measures: Following Federici et al. [22], we quantify each distribution shift by a mutual information term with respect to the environment variable E . Specifically, $I(\mathbf{X}; E)$, $I(Y; E)$, $I(\mathbf{X}; E | Y)$, $I(Y; E | \mathbf{X})$, and $I(\mathbf{X}, Y; E)$ measure, respectively, the shifts in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, and $\mathbb{P}(\mathbf{X}, Y)$ across environments; each vanishes exactly when the corresponding distribution is invariant. Existing OOD generalization methods typically assume that one of these terms is zero— $I(\mathbf{X}; E | Y) = 0$ under the label-shift assumption, or $I(Y; E | \mathbf{X}) = 0$ under the covariate-shift assumption. Under hidden confounding shift, by contrast, *both* terms are in general nonzero, so neither assumption holds and the corresponding guarantees no longer apply. This diagnostic—the simultaneous presence of several shift terms—is exactly what we test for on real-world data in § 5.

Challenges with hidden confounding shift: The experiment in Figure 2 makes the difficulty concrete. The outcome Y depends on an observed covariate X and a hidden confounder U , and environment-specific shifts in the mean of U , denoted μ_e , define distinct environments. A linear regression model trained on X alone recovers an *incorrect* X – Y relationship (first subplot): although Y increases with X within every environment, the pooled model fits the opposite trend—a textbook instance of Simpson’s paradox [73]. Tellingly, the same failure afflicts methods purpose-built for OOD generalization—IRM [6], VREX [38], and Group DRO [69]—which indicates that the failure is not particular to ERM but a structural consequence of confounding shift under minimal environment overlap. We prove in § 4 that the remedy is to learn environment-specific relationships, and the second subplot confirms it: augmenting the model with environment-specific summary statistics of X —mean, standard deviation, and quantiles—recovers the correct within-environment relationship, mirroring the *backdoor adjustment* criterion of causal-effect estimation [58], in which environment-specific information serves as a proxy for the confounder. Such statistics have limited representational capacity, however, so the environment-specific information encoded in the observed covariates alone can be insufficient. The third subplot shows the resolution developed in § 4.3: supplying additional informative covariates that act as proxies for the unobserved confounder yields a further improvement in generalization, approaching the oracle model trained directly on X and U , which sets the best achievable mean squared error.

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

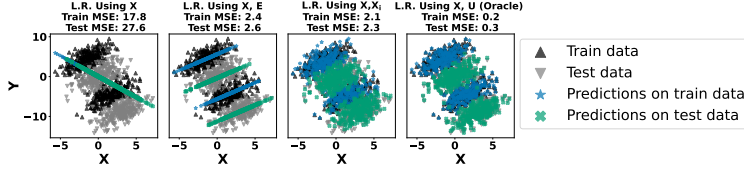


Figure 2: Four linear regression (L.R.) models evaluated in an OOD setting with hidden confounding shift and minimal environment overlap (distant μ_e), reported as MSE. (i) Trained on X alone, the model learns an incorrect X – Y relationship—an instance of Simpson’s paradox. (ii) Augmenting X with environment-specific summary statistics, denoted E , recovers the correct relationship, though its accuracy is bounded by the limited representational power of those statistics. (iii) Adding an informative covariate X_i for U improves OOD generalization further. (iv) The oracle model, trained on X and U , sets the best achievable performance. Performance improves monotonically from (i) to (iv).

4 Impact of hidden confounding shifts on OOD generalization

This section develops a theoretical account of how hidden confounding shift acts on the several facets of OOD generalization. Let $\hat{Y} = (f \circ \phi)(\mathbf{X})$ denote the predicted label for an input $\mathbf{X}$, where f is a classifier and ϕ a feature extractor (equivalently, a representation map). We write $H(X) = -\mathbb{E}_X[\log(\mathbb{P}(X))]$ for the entropy and $I(X; Y) = \mathbb{E}_{X,Y}[\log \frac{\mathbb{P}(X,Y)}{\mathbb{P}(X)\mathbb{P}(Y)}]$ for the mutual information, and denote by $\mathcal{R}^e(f \circ \phi) = \mathbb{E}_{(\mathbf{X},Y) \sim \mathbb{P}_{\mathbf{X},Y}^e}[\ell((f \circ \phi)(\mathbf{X}), Y)]$ the risk of a predictor $(f \circ \phi)$ in environment e under a loss $\ell(\cdot, \cdot)$. The goal of OOD generalization is to learn a predictor that performs well on both ID and OOD data, and the literature operationalizes this goal in several distinct ways. Robust-optimization methods [7, 9, 19, 75, 69] minimize the worst-case risk over training environments, $\mathcal{R}^{\text{Rob}}(f \circ \phi) = \max_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$, where $\mathcal{E}_{tr}$ is the set of training environments. Variance risk extrapolation (VREX) [38] minimizes $\mathcal{R}^{\text{VREX}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi) + \beta \times \text{Variance}(\{\mathcal{R}^1(f \circ \phi), \dots, \mathcal{R}^{|\mathcal{E}_{tr}|}(f \circ \phi)\})$, penalizing cross-environment risk variance through a hyperparameter β . Invariant risk minimization (IRM) [6] minimizes $\sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$ subject to the constraint that f be simultaneously optimal in every environment. Empirical risk minimization (ERM) [6, 38], by contrast, pools the data from all training environments and minimizes the empirical risk on the pool, $\mathcal{R}^{\text{ERM}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \sum_{(\mathbf{X},Y) \in \mathcal{D}^e} \ell((f \circ \phi)(\mathbf{X}), Y)$, where $\mathcal{D}^e$ is the observed data in environment e . These objectives differ chiefly in how—and whether—they constrain the representation to be invariant across environments, a distinction the decomposition below renders precise.

To expose how hidden confounding shift acts on these objectives, we adopt a unifying lens: the maximization of the predictive information $I(Y; \hat{Y})$, where $\hat{Y} = (f \circ \phi)(\mathbf{X})$. This quantity measures how informative the prediction $\hat{Y}$ is about the true label Y and therefore serves as a natural, task-agnostic objective for supervised learning, of which the risk-based objectives above can be seen as specific instantiations. We begin by formalizing the two properties of the representation $\phi(\mathbf{X})$ —*informativeness* and *invariance*—that jointly govern OOD generalization [93].

Definition 4.1 (Informativeness and Conditional Informativeness). *The informativeness of features $\phi(\mathbf{X})$ for predicting Y is defined as $\text{INF}(\phi(\mathbf{X}), Y) = I(\phi(\mathbf{X}); Y)$. The conditional informativeness of features $\phi(\mathbf{X})$ for predicting Y conditioned on environment variable E is defined as $\text{CINF}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); Y|E)$.*

Conditional informativeness quantifies the information $\phi(\mathbf{X})$ carries about Y within each environment. Minimizing CINF discards useful signal; maximizing it, however, is not universally desirable, since in settings such as algorithmic fairness $\phi(\mathbf{X})$ may absorb sensitive or biased within-environment information and yield unfair predictions. Such biased representations can be suppressed by minimizing $I(\phi(\mathbf{X}); E)$, but this can also curtail predictive performance, as we show in § 4.2. In the setting we study—where E carries no sensitive information—we adopt, and subsequently substantiate, the position that maximizing CINF improves generalization (§ 4.1).

Definition 4.2 (Variation and Invariance). *For a given label Y , the variation in the features $\phi(\mathbf{X})$ across environments E is defined as $\text{VAR}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); E|Y)$. For a given label Y , the invariance of the features $\phi(\mathbf{X})$ across environments E is defined as the negative of the variation i.e., $\text{INV}(\phi(\mathbf{X}), Y, E) = -\text{VAR}(\phi(\mathbf{X}), Y, E) = -I(\phi(\mathbf{X}); E|Y)$.*

Invariance measures how consistent the representation $\phi(\mathbf{X})$ is across environments for a given label Y . Minimizing invariance retains environment-specific information in $\phi(\mathbf{X})$ —and, in the absence of confounding, thereby invites overfitting—whereas maximizing it strips out environment-specific dependencies and enforces invariant learning. Extending the measures of § 3 to the learned representation, we quantify *feature shift* and *concept shift* by $I(\phi(\mathbf{X}); E)$ and $I(Y; E \mid \phi(\mathbf{X}))$, respectively; where the context is unambiguous, we let *concept shift* denote either $I(Y; E \mid \mathbf{X})$ or $I(Y; E \mid \phi(\mathbf{X}))$.

4.1 A general decomposition of predictive information

We first fix the requisite causal-graph terminology. A causal graph $\mathcal{G}$ comprises nodes, representing random variables, and directed edges, encoding direct causal influences. A *path* between two nodes X_i and X_j is a sequence of distinct nodes joined by edges; a *directed path* from X_i to X_j ($i < j$) is one in which every arrow points toward X_j , i.e. $X_i \rightarrow X_{i+1} \rightarrow \dots \rightarrow X_{j-1} \rightarrow X_j$. Along such a path, X_i is the *parent* of X_{i+1} , X_{i+1} the *child* of X_i , X_i an *ancestor* of X_j , and X_j a *descendant* of X_i . Every path decomposes into three elementary structures: a *chain* $X_i \rightarrow X_j \rightarrow X_k$, a *fork* $X_i \leftarrow X_j \rightarrow X_k$, and a *collider* $X_i \rightarrow X_j \leftarrow X_k$. In a chain or a fork, X_i and X_k are marginally dependent but rendered conditionally independent by the intermediate node X_j , i.e. $X_i \not\perp\!\!\!\perp X_k$ and $X_i \perp\!\!\!\perp X_k \mid X_j$. In a collider the relationship inverts: X_i and X_k are marginally independent yet become conditionally dependent once one conditions on X_j or any of its descendants, i.e. $X_i \perp\!\!\!\perp X_k$ and $X_i \not\perp\!\!\!\perp X_k \mid X_j$. A path between X_i and X_k is *blocked* by a conditioning set S if and only if (i) it contains a chain or fork whose middle node lies in S , or (ii) it contains a collider such that neither the collider nor any of its descendants belongs to S . When every path from X_i to X_k is blocked by S , we have $X_i \perp\!\!\!\perp X_k \mid S$; a path that is not blocked is *open*. This *d*-separation machinery is exactly what we invoke below to trace the information-flow consequences of conditioning under confounding shift.

We now turn to the predictive information $I(Y; \hat{Y})$, where the prediction $\hat{Y} = (f \circ \phi)(\mathbf{X})$ is formed from a learned representation $\phi(\mathbf{X})$. Figure 3 specifies the causal relationships among $\mathbf{X}, Y, U, \phi(\mathbf{X}), \hat{Y}$, and E , where U is a hidden confounder and E is the environment variable encoding shifts in $\mathbb{P}(U)$, so that any two environments differ precisely by a shift in $\mathbb{P}(U)$.

Because $\hat{Y}$ is a deterministic function of $\phi(\mathbf{X})$, it carries no information about Y beyond what $\mathbf{X}$ or $\phi(\mathbf{X})$ already provides: $\hat{Y} \perp\!\!\!\perp Y \mid \mathbf{X}$ and $\hat{Y} \perp\!\!\!\perp Y \mid \phi(\mathbf{X})$. We first derive a general

decomposition of the predictive information that makes no commitment as to how U influences $\mathbf{X}$ and Y , and defer the treatment of that influence to § 4.2.

Proposition 4.1. *For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some covariates cause Y and some covariates are caused by Y , environment variable E , a feature extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:*

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y \mid E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E \mid Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E \mid \phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y \mid \hat{Y})}^{\text{Residual}} \quad (1)$$

where $I(\phi(\mathbf{X}); Y \mid \hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

Proofs are deferred to Appendix § A. Beyond its role in the subsequent analysis, the decomposition offers a unifying vantage point on existing methods, many of which can be read as explicitly maximizing or minimizing particular terms of Equation (1). IRM [6], by constraining a fixed classifier f to remain optimal across environments, effectively maximizes $-I(\phi(\mathbf{X}); E \mid Y)$, since any residual environment-specific information in $\phi(\mathbf{X})$ conditional on Y would break the classifier’s cross-environment optimality. DANN [23] and the independence criterion for fair classification [22] minimize the environment-specific information $I(\phi(\mathbf{X}); E)$, and several domain-adaptation methods likewise learn features whose statistics are invariant to the domain [82, 90]. Minimizing $I(\phi(\mathbf{X}); E)$, though beneficial for fair classification, can degrade predictive performance [98, 33, 22]—a trade-off that Equation (1) renders transparent, since reducing $I(\phi(\mathbf{X}); E)$ lowers $I(Y; \hat{Y})$ term-for-term. CDAN [51] targets $\mathbb{P}^e(\phi(\mathbf{X}), Y) = \mathbb{P}^{e'}(\phi(\mathbf{X}), Y)$; enforcing this equality makes E independent

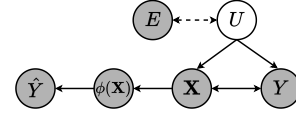


Figure 3: The bi-directed arrow between $\mathbf{X}$ and Y indicates that some covariates in $\mathbf{X}$ cause Y while others are caused by Y .

of the pair $(\phi(\mathbf{X}), Y)$, whence $\mathbb{P}(Y \mid \phi(\mathbf{X}), E) = \mathbb{P}(Y \mid \phi(\mathbf{X})) \implies I(Y; E \mid \phi(\mathbf{X})) = 0$, so that although CDAN’s loss does not name $I(Y; E \mid \phi(\mathbf{X}))$, its joint-distribution alignment drives this conditional mutual information to zero. Finally, while $I(Y; E)$ is constant with respect to the representation, dedicated methods exist that are robust to shifts in the label distribution [69, 95]. We stress that these methods differ in *how strongly* they act on their target terms—from hard constraints to soft penalties—so the decomposition indicates which term a method emphasizes rather than asserting that any method eliminates it outright.

Equation (1) was obtained without committing to how U influences $\mathbf{X}$ and Y ; specifying that influence sharpens the guidance for handling its individual terms. If $U \rightarrow Y$, $U \not\rightarrow \mathbf{X}$, $Y \rightarrow \mathbf{X}$, then $I(\mathbf{X}; E \mid Y) = 0$, identifying $I(\phi(\mathbf{X}); E \mid Y)$ as the principled quantity to minimize. Symmetrically, if $U \not\rightarrow Y$, $U \rightarrow \mathbf{X}$, $\mathbf{X} \rightarrow Y$, then $I(Y; E \mid \mathbf{X}) = 0$, motivating the minimization of $I(Y; E \mid \phi(\mathbf{X}))$. The genuinely difficult—and, we argue, the practically prevalent—regime is the one in which $U \rightarrow Y$ and $U \rightarrow \mathbf{X}$: the terms of Equation (1) then interact nontrivially, and no single term can be identified in advance as the one to control. The next section resolves precisely this case.

4.2 Predictive information under hidden confounding shift

To determine how a hidden confounder U that causes both $\mathbf{X}$ and Y shapes the predictive information, we analyze the *two exhaustive cases* for the $\mathbf{X}$ – Y relationship: (i) $\mathbf{X} \rightarrow Y$ and (ii) $Y \rightarrow \mathbf{X}$. Under $\mathbf{X} \rightarrow Y$, the inequalities in (2) follow directly from the d -separation rules of § 4.1: conditioning on Y opens the path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ at the collider node Y , admitting additional information flow from $\phi(\mathbf{X})$ to E through Y ; conversely, conditioning on $\phi(\mathbf{X})$ partially blocks the flow from Y to E at $\mathbf{X}$, to the extent that $\phi(\mathbf{X})$ retains information about $\mathbf{X}$.

$$(i) \overbrace{I(\phi(\mathbf{X}); E \mid Y)}^{\text{Variation}} \geq \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \quad (ii) \overbrace{I(Y; E)}^{\text{Label shift}} \geq \overbrace{I(Y; E \mid \phi(\mathbf{X}))}^{\text{Concept shift}} \quad (2)$$

Symmetrically, under $Y \rightarrow \mathbf{X}$, the inequalities in (3) hold: conditioning on Y now blocks the flow from $\phi(\mathbf{X})$ to E at Y , while conditioning on $\phi(\mathbf{X})$ opens the path $Y \rightarrow \mathbf{X} \leftarrow U \leftarrow E$ from Y to E at $\mathbf{X}$ —because $\phi(\mathbf{X})$ is a child of the collider $\mathbf{X}$, and conditioning on the child of a collider opens the path through it [58] (cf. the preliminaries in § 4.1).

$$(i) \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \geq \overbrace{I(\phi(\mathbf{X}); E \mid Y)}^{\text{Variation}} \quad (ii) \overbrace{I(Y; E \mid \phi(\mathbf{X}))}^{\text{Concept shift}} \geq \overbrace{I(Y; E)}^{\text{Label shift}} \quad (3)$$

These inequalities, (2) and (3), allow the general decomposition to collapse into a markedly sharper form.

Proposition 4.2. *For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:*

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y \mid E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y \mid \hat{Y})}^{\text{Residual}} \quad (4)$$

Proposition 4.2 treats the two cases $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$ separately, and although both are covered, the case $\mathbf{X} \rightarrow Y$ is the one that predominates in real-world tabular prediction: 11 of the 16 datasets surveyed in recent benchmark studies [57, 50, 24] follow this structure, and in the remaining 5 the covariates caused by Y are far outnumbered by those that cause Y . The proposition’s central consequence is stated below.

Relative to the general decomposition in (1), the confounded decomposition in (4)—valid in the cases (i) $U \rightarrow \mathbf{X}, U \rightarrow Y, \mathbf{X} \rightarrow Y$ and (ii) $U \rightarrow \mathbf{X}, U \rightarrow Y, Y \rightarrow \mathbf{X}$ —is entirely free of the *variation, label shift, feature shift, and concept shift* terms. Consequently, under hidden confounding shift, whenever $\mathbf{X} \rightarrow Y$ or $Y \rightarrow \mathbf{X}$, maximizing predictive information reduces *exactly* to maximizing the single difference *conditional informativeness* – *residual*—the environment-specific objective that invariance-only methods, by construction, leave undressed.

Maximizing conditional informativeness: Prashant et al. [62] recently proposed a mixture-of-experts (MoE) model for OOD generalization under hidden confounding shift, in which each expert

is tied to a specific hidden-confounder assignment and therefore maximizes performance within the environment corresponding to that confounder value. Training such an MoE is thus an instance of maximizing the conditional informativeness $I(\phi(\mathbf{X}); Y | E)$, and Equation (4) supplies a theoretical justification for this class of models. The same decomposition also delineates the limits of existing instantiations and the room for methods that operate under more general confounding shift: the approach of Prashant et al. [62] presupposes overlapping confounder support, $\text{supp}(\mathbb{P}(U | \mathcal{E}_{te})) \subseteq \text{supp}(\mathbb{P}(U | \mathcal{E}_{tr}))$, together with discrete-valued proxies.

Figure 2 shows that, even in a setting *without* confounder overlap and with a *continuous-valued* proxy—conditions that fall outside the assumptions above—a linear regression model equipped with environment-specific information recovers the correct causal relationship between the observed covariates and the target across environments. This example makes concrete the need for OOD generalization methods that remain effective when confounder support is limited and the confounder is continuous-valued.

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Minimizing residual: The residual term $I(\phi(\mathbf{X}); Y | \hat{Y})$ measures the information $\phi(\mathbf{X})$ retains about Y beyond what the prediction $\hat{Y}$ already conveys, and it admits the expansion $I(\phi(\mathbf{X}); Y | \hat{Y}) = H(Y | \hat{Y}) - H(Y | \phi(\mathbf{X}), \hat{Y})$. The conditional entropy $H(Y | \hat{Y})$ connects directly to the expected cross-entropy loss through $\mathbb{E}_{Y, \hat{Y}}[\ell_{\text{CE}}(Y, \hat{Y})] = H(Y | \hat{Y}) + \text{calibration error}$ [12, 10, 29], in which $H(Y | \hat{Y})$ —the *refinement error*—captures the model’s discriminative power over classes. Two complementary strategies for minimizing the residual therefore follow. First, one can drive $I(\phi(\mathbf{X}); Y | \hat{Y})$ to zero directly by rendering $\phi(\mathbf{X})$ conditionally independent of the target given the prediction, $Y \perp\!\!\!\perp \phi(\mathbf{X}) | \hat{Y}$. Alternatively, one can reduce the refinement error $H(Y | \hat{Y})$ —for instance, through training schemes that first minimize refinement error to secure accurate predictions and only then address calibration error [10].

352 4.3 Impact of additional informative variables

Section 4.2 characterized the optimal generalization strategy when the confounder is entirely unobserved. In practice, however, proxies for the hidden confounder are frequently available [4, 85, 62] and can be enlisted to stand in for it. We now quantify the effect that admitting such non-causal but informative covariates has on the predictive information.

Definition 4.3 (Informative Covariates). *A set of covariates $\mathbf{X}_I$ that are not causally related to Y i.e., neither ancestors nor descendants of Y are in $\mathbf{X}_I$, are said to be informative to Y if $\mathbf{X}_I$ and Y are not independent of each other given other causally related covariates $\mathbf{X}$ and E i.e., $Y \not\perp\!\!\!\perp \mathbf{X}_I | \mathbf{X}, E$.*

Because $U \rightarrow Y$, any covariate informative about U is also informative about Y , and conversely. The decomposition in (1) shows that maximizing $I(Y; \hat{Y})$ requires, among other movements, minimizing the concept shift $I(Y; E | \phi(\mathbf{X}))$. This is consistent with the empirical finding of Liu et al. [50] that collecting additional covariates $\mathbf{X}_I$ which render $\mathbb{P}(Y | \mathbf{X}, \mathbf{X}_I)$ more stable across environments—that is, which reduce concept shift—improves OOD test accuracy. The following proposition places that observation on a firm theoretical footing, establishing that admitting more informative variables increases predictive information by moving the relevant terms of (1) in their beneficial directions.

Proposition 4.3. *If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that $Y \not\perp\!\!\!\perp \mathbf{X}_I | \mathbf{X}, E$, then we have the following inequalities:*

$$\begin{aligned} (i) \quad & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y | E) > I(\phi_1(\mathbf{X}); Y | E) \quad (ii) \quad I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) \quad & I(Y; E | \phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E | \phi_1(\mathbf{X})) \quad (iv) \quad I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y) \end{aligned} \quad (5)$$

Proposition 4.3 establishes that admitting informative covariates increases conditional informativeness (5.i) and feature shift (5.ii) while reducing concept shift (5.iii)—three movements that are each favorable for predictive information. The same covariates, however, also amplify variation (5.iv); realizing their full benefit may therefore call for dedicated strategies to control this competing term, a point we return to in the discussion.

Table 2: Distribution shifts quantified by the mutual information between each quantity and the environment variable. Each entry is the mean $\pm$ standard deviation over 10 random subsets of 40,000 samples; one-sample t -tests against a mean of zero ($\alpha = 0.05$) find every measure significantly nonzero (p-value ≈ 0). All four shift types thus co-occur in every dataset—the observable signature that hidden confounding shift necessarily produces (§ 3).

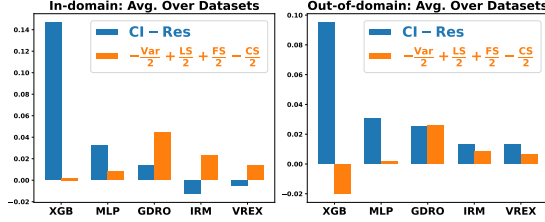
Dataset	Conditional Covariate Shift $I(\mathbf{X}; E Y)$	Label shift $I(Y; E)$	Covariate Shift $I(\mathbf{X}; E)$	Concept shift $I(Y; E \mathbf{X})$
Readmission	0.107 ± 0.002	0.068 ± 0.002	0.097 ± 0.002	2.032 ± 0.000
Food stamps	0.126 ± 0.004	0.030 ± 0.003	0.108 ± 0.001	2.118 ± 0.001
Income	0.168 ± 0.002	0.075 ± 0.003	0.147 ± 0.001	2.059 ± 0.002
Public coverage	0.231 ± 0.002	0.412 ± 0.006	0.222 ± 0.002	1.945 ± 0.001
Unemployment	0.117 ± 0.001	0.019 ± 0.002	0.114 ± 0.002	2.010 ± 0.003
Diabetes	0.032 ± 0.002	0.048 ± 0.001	0.022 ± 0.002	2.132 ± 0.001
Hypertension	0.090 ± 0.002	0.183 ± 0.003	0.037 ± 0.001	1.883 ± 0.004
ASSISTments	0.293 ± 0.002	0.260 ± 0.001	0.306 ± 0.002	0.367 ± 0.004

5 Experimental results

Our experiments on real-world and synthetic datasets are designed to examine three claims: (i) the statistical signature of hidden confounding shift is present in real-world data; (ii) the components of the decomposition in (1) relate to OOD generalization under hidden confounding shift in the manner the theory predicts; and (iii) informative covariates improve generalization. The evidence on the real-world benchmarks is observational and hence associational; we complement it with fully controlled synthetic studies (§ D) in which the confounding structure is known by construction, so that the two lines of evidence together isolate the mechanism the theory identifies. We evaluate on eight real-world tabular benchmark datasets—*Food stamps*, *Readmission*, *Income*, *Public coverage*, *Unemployment*, *Diabetes*, *Hypertension*, and *ASSISTments*—together with their domain splits, all drawn from the *TableShift* benchmark [24], and we report ID and OOD test accuracy. The comparison spans five methods representing the principal OOD paradigms: two ERM-based methods, XGBoost (XGB) [15] and a multi-layer perceptron (MLP); two domain-generalization methods, IRM [6] and VREX [38]; and one robust-learning method, Group DRO (GDRO) [69]. Reported values are averages over 5 random hyperparameter draws (Appendix B); the effect of random seeds is statistically insignificant on these datasets [24]. Full setup details appear in Appendix B, and results under explicit hyperparameter tuning in Appendix C.

Hidden confounding shift: As established in § 3, a hidden confounding shift *implies* the simultaneous presence of observable shifts in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y)$, and $\mathbb{P}(Y | \mathbf{X})$; the co-occurrence of all four is therefore a necessary observable signature of hidden confounding, though not, on its own, proof that a shifting common cause is the only possible explanation. Table 2 shows that this signature is present throughout: label, covariate, conditional-covariate, and concept shift are each significantly nonzero across every real-world dataset, a pattern consistent with pervasive hidden confounding shift and difficult to reconcile with any single pure-shift assumption. As a purely qualitative complement—which we deliberately keep separate from this quantitative evidence—we query GPT-4o [1] to enumerate *plausible* candidate confounders for several benchmark datasets [24, 57] (Appendix E); for the *food stamps* dataset, for instance, unmeasured factors such as state-specific *economic policies* could plausibly influence both *household income* and *food stamp reciprocity*. These generated candidates are illustrative and not independently verified, and we treat them as hypotheses that motivate the mechanism rather than as evidence for it.

Conditional informativeness vs. accuracy: Proposition 4.2 predicts that, under hidden confounding shift, maximizing predictive information reduces to maximizing the difference *conditional informativeness* – *residual*. Figure 4 examines this prediction on the six datasets of Table 2 that contain more than one training environment [57, 24] (per-dataset results in Appendix C). The observed pattern matches the prediction: across methods, the difference *conditional informativeness* – *residual* is positively associated with both ID and OOD accuracy, and XGB—which attains the largest value of this difference—is also the best-performing model on both metrics. In line with the theory, the per-dataset results in Appendix C show the *orange* bars collapsing to near zero, since under hidden confounding shift the sum $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ vanishes (Proposition 4.2). The relative ID/OOD ordering of the methods reproduces the rankings previously reported in the literature [57, 24], indicating that our eval-



Method	Avg. ID Test Acc.	Avg. OOD Test Acc.
XGB	85.22	77.00
MLP	83.64	73.29
GDRO	78.40	72.28
IRM	71.17	64.89
VREX	59.74	63.06

Figure 4: The difference *conditional informativeness* – *residual* (left) tracks the average ID and OOD test accuracy (right) across the eight datasets: ranking the five methods by this difference recovers their ranking by accuracy, with XGB leading on both ID (85.22%) and OOD (77.00%) accuracy. The remaining decomposition terms (orange) contribute negligibly, as predicted for the confounded regime.

Table 3: Sign consistency of each decomposition term with accuracy gains, alongside ID and OOD test accuracy across the causal (C), arguably causal (AC), and all (A) covariate settings. Concept shift (CS) is highly sign-consistent for every method; the term that Proposition 4.2 identifies as governing generalization, however, is conditional informativeness (CI), and it is on this term that XGB attains both the highest sign consistency (0.89) and the highest accuracy in every column. Res: residual, Var: variation, FS: feature shift.

	Sign Consistency Metric ($\uparrow$)					ID-Test Accuracy ($\uparrow$)			OOD-Test Accuracy ($\uparrow$)		
Method	CI	Var	FS	CS	Res	C	AC	A	C	AC	A
XGB	0.89	0.61	0.36	0.89	0.17	81.24	84.84	85.22	69.92	76.72	77.00
MLP	0.75	0.64	0.28	0.89	0.33	80.51	81.93	83.64	69.70	73.84	73.29
GDRO	0.69	0.67	0.19	0.94	0.36	80.92	83.76	78.40	69.61	73.49	72.28
IRM	0.78	0.61	0.25	0.92	0.25	67.91	68.22	71.17	61.21	62.96	64.89
VREX	0.64	0.61	0.25	0.89	0.44	54.39	62.32	59.74	56.62	63.53	63.06

uation is consistent with established benchmarks. As with any observational analysis, this evidence is associational; the controlled synthetic study in § D supplies the complementary interventional check.

Informative covariates vs. accuracy: We next isolate the contribution of informative covariates to OOD generalization. Adopting the covariate partition of Nasti and Hardt [57], we organize all available covariates into three nested subsets, *causal covariates* $\subseteq$ *arguably causal covariates* $\subseteq$ *all covariates* (Appendix § B), and measure how the terms of the decomposition in (1) respond as the subset is enlarged. To make this response legible, we introduce the *sign consistency metric*: upon each enlargement of the covariate set, a positively weighted term (e.g. conditional informativeness) is scored as consistent when its value rises together with an accuracy gain, and a negatively weighted term (e.g. concept shift) when its value falls; the metric reports the fraction of comparisons in which a term moves in its *beneficial* direction, and thus measures how reliably that term co-moves with generalization (formal definition in Appendix § B). Table 3 shows that both conditional informativeness and concept shift are highly sign-consistent with accuracy gains. Reducing concept shift is common to all five methods; what sets XGB apart is that it additionally *raises* conditional informativeness—the term the theory singles out as decisive—which coincides with its superior accuracy in every covariate setting. Dataset-specific results appear in Appendix § C.

On real-world data one cannot verify whether the *all available covariates* set in fact contains *every* relevant or sufficiently informative variable. To examine the hypothesis under fully controlled conditions, we therefore turn to synthetic data with the known causal structure $U \rightarrow X$, $U \rightarrow Y$, $X \rightarrow Y$, and $U \rightarrow X_I$, where X_I denotes informative covariates generated by the hidden confounder U . Here the confounding is present by construction, so the analysis is interventional rather than merely associational. As Figure 5 shows, introducing X_I increases conditional informativeness and feature shift and reduces concept shift—exactly the three movements Proposition 4.3 predicts—while the variation term remains at zero owing to the simplicity of this design. The data-generating process and further results are given in Appendix § D.

6 Discussion, limitations, and future work

We have decomposed the predictive information between model outputs and true outcomes into a small set of interpretable terms that expose the factors limiting OOD generalization (Equation (1)). This decomposition yields a coherent and unifying explanation for an otherwise puzzling body of empirical results: when a hidden confounder shifts across environments—registered as coupled changes in the covariate and outcome distributions—no finite set of observed covariates or proxies can fully

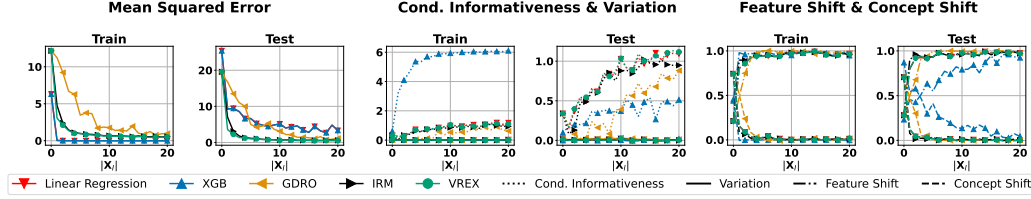


Figure 5: On synthetic data with known confounding structure, adding proxy variables $\mathbf{X}_I$ for U that are informative about Y monotonically reduces MSE while increasing conditional informativeness and feature shift and decreasing concept shift—precisely the three movements predicted by Proposition 4.3, with the variation term identically zero in this design.

dissolve the ambiguity it induces. Because such a shift entangles covariate, label, concept, and feature shifts (the inequalities in (2) and (3)), a strategy that addresses these shifts in isolation is unlikely to suffice. Our central theoretical result (Equation (4)) proves that, in this regime, maximizing predictive information reduces to learning environment-specific input–output mappings, so that enforcing a single invariant predictor is provably insufficient on its own; our experiments across real-world and synthetic data are consistent with this conclusion. These results position hidden confounding shift as a mechanism that robust OOD methods should confront directly, and they establish that it is at once analytically characterizable and, given informative proxies, tractable.

Several limitations delimit these conclusions and mark the natural next steps. First, the theory rests on structural assumptions—about the causal graph, the feature extractor, the relevant conditional independences, and information preservation under ϕ —whose robustness to violation on real data warrants further study. Second, the real-world evidence is observational and confined to tabular benchmarks with a compact set of methods; the connection between the decomposition terms and accuracy is associational, and the mutual-information estimates, while stable across sample sizes in our checks (Appendix § B), remain subject to the usual estimation caveats in mixed discrete–continuous settings. Third, the analysis presumes the modality-specific notion of observed proxy covariates and causal feature groups that tabular data affords; extending it to image, language, and multimodal settings, where these notions are less explicit, is an open direction. Building on the framework, the most consequential open problems are: (i) algorithms that remain effective under limited confounder overlap; (ii) a quantitative account of the cost–accuracy trade-off in acquiring informative covariates, and of how a practitioner might choose among ERM, invariant learning, and proxy enrichment without access to test environments; (iii) methods that handle entangled shifts without recourse to untestable proxy assumptions; and (iv) explicit modeling of the ambiguity introduced by unobserved confounders, as a route to genuinely new OOD-robust paradigms.

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748 Appendix

749 A Proofs of propositions

750 **Proposition 4.1.** For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some
 751 covariates cause Y and some covariates are caused by Y , environment variable E , a feature
 752 extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E|Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E|\phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

753 where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the
 754 prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

755 *Proof.* From the causal graph shown in Figure 3 of the main paper, in the causal substructure:
 756 $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$, $\hat{Y}$ provides no additional information in predicting Y given either $\phi(\mathbf{X})$ or
 757 $\mathbf{X}$. This is because, the path $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$, between $\hat{Y}$ and Y is blocked by either $\{\phi(\mathbf{X})\}$
 758 or $\{\mathbf{X}\}$. This implies $\hat{Y} \perp\!\!\!\perp Y|\phi(\mathbf{X})$ and $\hat{Y} \perp\!\!\!\perp Y|\mathbf{X}$. Now consider the expansion of $I(Y; \hat{Y})$.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}), E; Y) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (6)$$

759 Similarly,

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(Y, E; \phi(\mathbf{X})) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (7)$$

760 Summing Equations (6) and (7) and grouping the terms gives the desired expression:

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); Y|E) - \frac{I(\phi(\mathbf{X}); E|Y)}{2} + \frac{I(Y; E)}{2} \\ &\quad + \frac{I(\phi(\mathbf{X}); E)}{2} - \frac{I(Y; E|\phi(\mathbf{X}))}{2} - I(\phi(\mathbf{X}); Y|\hat{Y}). \end{aligned}$$

761 To get the similar decomposition with $\mathbf{X}$ instead of $\phi(\mathbf{X})$, use $\hat{Y} \perp\!\!\!\perp Y | \mathbf{X}$ and introduce $\mathbf{X}$ into the
 762 expansion instead of $\phi(\mathbf{X})$. $\square$

763 **Proposition 4.2.** For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor
 764 ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if
 765 either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

766 *Proof.* Following the causal graph shown in Figure 3 of the main paper, we prove the proposition in
 767 two separate cases: $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$.

768 **Case 1: $\mathbf{X} \rightarrow Y$.** In this case, we have the following inequality because conditioning on Y opens the
 769 path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ between $\phi(\mathbf{X})$ and E at the collider node Y :

$$I(\phi(\mathbf{X}); E|Y) \geq I(\phi(\mathbf{X}); E). \quad (8)$$

770 Additionally, considering the causal substructure: $Y \leftarrow \mathbf{X} \leftarrow U \leftrightarrow E$, we have the following
 771 inequality because conditioning on $\phi(\mathbf{X})$ partially blocks the information flow from Y to E at the
 772 node $\mathbf{X}$ as long as $\phi(\mathbf{X})$ encodes some information about $\mathbf{X}$:

$$I(Y; E) \geq I(Y; E|\phi(\mathbf{X})). \quad (9)$$

773 Now consider the following expansion from Equation (7) in the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (8)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

774 Similarly, consider the following expansion from Equation (6) from the proof of Proposition 4.1:

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\geq I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (9)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

775 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

776 **Case 2: $Y \rightarrow \mathbf{X}$.** In this scenario, we have the following inequality because conditioning on $\phi(\mathbf{X})$
 777 opens the path $E \rightarrow U \rightarrow \mathbf{X} \leftarrow Y$ between Y and E at the collider node $\mathbf{X}$ because $\phi(\mathbf{X})$ is a child
 778 of $\mathbf{X}$ [58].

$$I(Y; E|\phi(\mathbf{X})) \geq I(Y; E) \quad (10)$$

779 Additionally, we have the following inequality because conditioning on Y blocks the path $\phi(\mathbf{X}) \leftarrow$
 780 $\mathbf{X} \leftarrow Y \leftarrow U \leftrightarrow E$ at Y and hence less information flow between $\phi(\mathbf{X})$ and E .

$$I(\phi(\mathbf{X}); E) \geq I(\phi(\mathbf{X}); E|Y) \quad (11)$$

781 Now consider the following expansion from Equation (6) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(Y; E|\phi(\mathbf{X})) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (10)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

782 Similarly, consider the following expansion from Equation (7) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\geq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (11)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

783 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

784 Since the upper bound and lower bounds are the same in both cases, we have the desired result
 785 $I(Y; \hat{Y}) = I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$. $\square$

786 **Proposition 4.3.** If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that
 787 $Y \not\perp \mathbf{X}_I \mid \mathbf{X}, E$, then we have the following inequalities:

$$\begin{aligned} (i) & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E) \quad (ii) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) & I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E|\phi_1(\mathbf{X})) \quad (iv) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) > I(\phi_1(\mathbf{X}); E \mid Y) \end{aligned} \quad (5)$$

788 *Proof.* (i) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; Y|E) = I(\mathbf{X}; Y|E) +$
789 $I(\mathbf{X}_I; Y|E, \mathbf{X})$. We have $I(\mathbf{X}_I; Y|E, \mathbf{X}) > 0$ because $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$. Hence, we have
790 $I(\mathbf{X}, \mathbf{X}_I; Y|E) > I(\mathbf{X}; Y|E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
791 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E)$.

792 (ii) We consider the causal graph shown in Figure 3. From the chain rule of mutual information,
793 we have: $I(\mathbf{X}, \mathbf{X}_I; E) = I(\mathbf{X}; E) + I(\mathbf{X}_I; E|\mathbf{X})$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is
794 informative to Y is also informative to U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}) > 0$, and hence
795 $I(\mathbf{X}, \mathbf{X}_I; E) > I(\mathbf{X}; E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
796 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E)$.

797 (iii) Consider $I(Y; E, \mathbf{X}_I | \mathbf{X})$. It can be expressed in two ways:

$$I(Y; E, \mathbf{X}_I | \mathbf{X}) = I(Y; E | \mathbf{X}) + I(Y; \mathbf{X}_I | E, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) + I(Y; E | \mathbf{X}_I, \mathbf{X})$$

798 Equating and rearranging the terms gives the following.

$$I(Y; E | \mathbf{X}) - I(Y; E | \mathbf{X}_I, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) - I(Y; \mathbf{X}_I | E, \mathbf{X})$$

799 Since $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid E, \mathbf{X}$, we have $I(Y; \mathbf{X}_I | \mathbf{X}) > I(Y; \mathbf{X}_I | E, \mathbf{X})$ because conditioning usually
800 reduces mutual information unless the additional conditioning variable opens any collider paths. Here
801 E is neither a collider nor a descendant of any collider. Thus, $I(Y; E | \mathbf{X}) > I(Y; E | \mathbf{X}_I, \mathbf{X})$. Since
802 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
803 we have $I(Y; E | \phi_1(\mathbf{X})) > I(Y; E | \phi_2(\{\mathbf{X}_I \cup \mathbf{X}\}))$.

804 (iv) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; E | Y) = I(\mathbf{X}; E | Y) +$
805 $I(\mathbf{X}_I; E|\mathbf{X}, Y)$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is informative to Y is also informative to
806 U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}, Y) > 0$, and hence $I(\mathbf{X}, \mathbf{X}_I; E | Y) > I(\mathbf{X}; E | Y)$. Since
807 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
808 we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y)$. $\square$

809 B Experimental setup

810 This section details our experimental setup. Because the impact of random seeds is statistically
811 insignificant in most settings [24], the main-paper results report the mean and standard deviation
812 over five random hyperparameter configurations—sampled from the ranges in Table B1—for each
813 dataset–method combination. Every mutual information term is estimated with the nonparametric
814 entropy-estimation toolbox [37, 77, 78], a k -nearest-neighbor estimator suited to the mixed discrete–continuous tabular features considered here, and, following Gardner et al. [24], each term is
815 evaluated separately on in-distribution (ID) test data drawn from the training domains and out-of-
816 distribution (OOD) test data drawn from the held-out test domains.
817

818 For consistency, we draw 20,000 random samples
819 from each of the ID and OOD test sets, except
820 where a dataset contains fewer than 20,000
821 samples. Sample size exerts a negligible effect
822 on the estimated mutual information terms
823 throughout: as Figure B1 illustrates, the terms
824 of Equation (1) computed for the XGB representations on the *income* dataset are essentially
825 invariant to it—a stability check that partially addresses the estimator-reliability concern, though
826 it does not eliminate the estimation bias inherent to high-dimensional mutual-information estimation.
827 The representation $\phi(\mathbf{X})$ is extracted from the information estimates preceding the classification
828 head; for XGBoost, it is formed by concatenating SHAP model output margins with its SHAP values.
829 All experiments were run on a single NVIDIA RTX A6000 GPU.

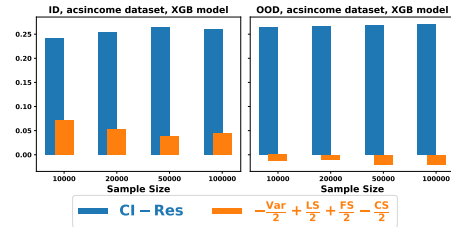


Figure B1: The evaluated predictive-information terms remain stable across sample sizes on the *income* dataset, indicating that the mutual information estimates preceding the classification head are stable. The information estimates preceding the classification head are stable. The information estimates preceding the classification head are stable.

833 For the experiments on the impact of informative covariates, we adopt the covariate categorization of Nastl and Hardt [57], which partitions the covariates into *causal*, *arguably causal*, and *all covariates*. Causal covariates have a clearly established influence on the target, with well-supported directionality and negligible risk of reverse causation; restricting a model to them nonetheless risks omitting genuine parents that remain unidentified. Arguably causal covariates

Table B1: Hyperparameters and their candidate search values for each method considered in this paper.

Model	Hyperparameter	Search values
XGBoost	Learning rate	[0.1, 0.2, 0.3]
	Maximum tree depth	[4, 5, 6]
	Minimum child weight	[0.1, 1, 10]
	Gamma (min. loss reduction)	[0.0001, 0.001, 0.01]
	Subsample ratio	[0.5, 0.6, 0.7]
	Column subsample per tree	[0.5, 0.6, 0.7]
	L1 regularization (α)	[0.0001, 0.001, 0.01]
	L2 regularization (λ)	[0.0001, 0.001, 0.01]
MLP	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
	Number of epochs	[1, 2, 3]
GroupDRO	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Group-weights step size	[0.01, 0.05, 0.1]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
	Number of epochs	[1, 2, 3]
IRM	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	IRM penalty weight (λ)	[0.01, 0.05, 0.1]
	IRM penalty anneal iterations	[1, 2, 3]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
	Number of epochs	[1, 2, 3]

838 carry uncertain causal status, satisfying at least one of: (1) a known causal effect, (2) a plausible
839 but possibly bidirectional influence, or (3) a likely but unconfirmed causal effect. Together,
840 these two groups approximate the true causal parents as closely as current knowledge permits,
841 although the underlying relationships may still be confounded. Some datasets additionally contain
842 *anti-causal* covariates, which the target likely influences rather than the reverse, and the *all*
843 *covariates* set comprises every observed variable irrespective of causal status. The sign consistency
844 metric introduced in § 5 is defined formally as follows. For each $(m, \sigma_m) \in M$, where
845 $M = \{(\text{conditional informativeness}, +1), (\text{variation}, -1), (\text{label shift}, +1), (\text{feature shift}, +1),$

846 (concept shift, -1), and (residual, -1)}, and for a given dataset, we compute:

$$C_m = \frac{1}{2|P|} \sum_{(s,t) \in P} [\mathbb{I}(\sigma_m \cdot (\mu_t^{\text{tr}}(m) - \mu_s^{\text{tr}}(m)) > 0) + \mathbb{I}(\sigma_m \cdot (\mu_t^{\text{te}}(m) - \mu_s^{\text{te}}(m)) > 0)]$$

847 Here P is the set of covariate-set pairs (s, t) with $t \supset s$ —for example, s the causal and t the arguably
848 causal covariates—and the factor of 2 in the denominator $2|P|$ accounts for the two data splits (train
849 and test). The term $\mu_s^{\text{tr/te}}(m)$ denotes the mean of measure m for covariate set s on the training/testing
850 data, $\mathbb{I}(\cdot)$ is the indicator function, and $|P| = 3$ for the settings $\{\text{causal}, \text{arguably causal}, \text{all}\}$. The
851 reported sign consistency for each m is the cross-dataset average $\bar{C}_m = \frac{1}{|D|} \sum_{d \in D} C_m^{(d)}$, in which
852 $|D|$ is the number of datasets.

853 C Additional results on real-world datasets

854 This section reports additional results on the real-world datasets. Figures C2 and C3 compare, per
855 dataset, the two composite quantities *conditional informativeness* $-$ *residual* and $-$ *variation* $/ 2 +$
856 *label shift* $/ 2 +$ *feature shift* $/ 2 -$ *concept shift* $/ 2$, whose aggregate appears in Figure 4 (left) of
857 the main paper. Two observations emerge. First, consistent with Proposition 4.1, the sum of the
858 blue and orange terms is positively associated with overall model performance. Second, under
859 the hidden confounding shift present in real-world data, the *conditional informativeness* $-$ *residual*
860 difference contributes substantially more to the total predictive information than does $-$ *variation* $/ 2 +$
861 *label shift* $/ 2 +$ *feature shift* $/ 2 -$ *concept shift* $/ 2$. Together these observations support maximizing
862 conditional informativeness as the primary lever for OOD generalization under hidden confounding
863 shift.

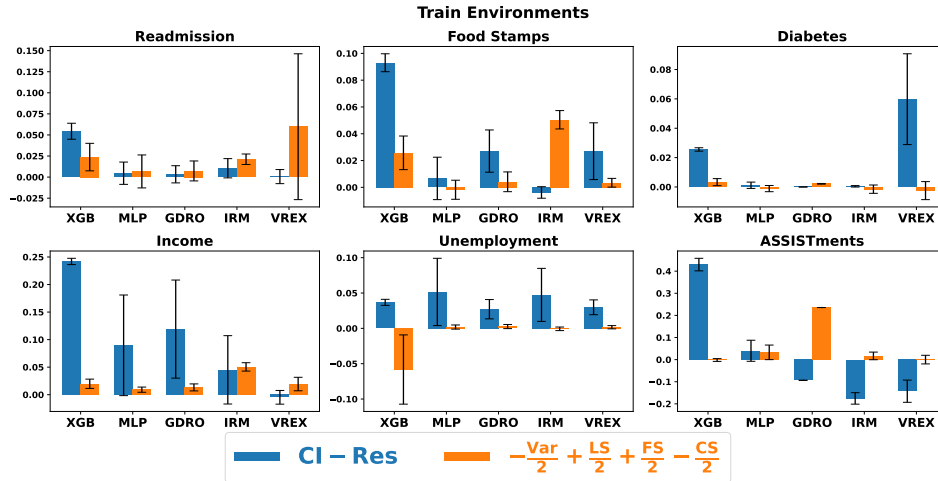


Figure C2: Dataset-specific decomposition on ID (train-domain) data, per method.

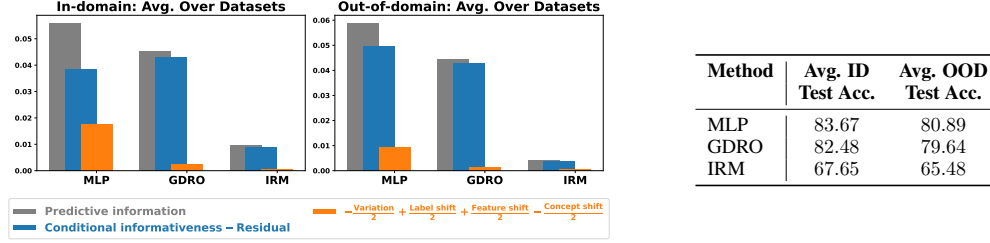


Figure C4: Under explicit hyperparameter tuning, the difference *conditional informativeness* – *residual* (left) remains positively associated with the average ID and OOD test accuracy over the five datasets (right) and continues to contribute more to the total predictive information than $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$, indicating that the main-paper conclusions are robust to the model-selection procedure.

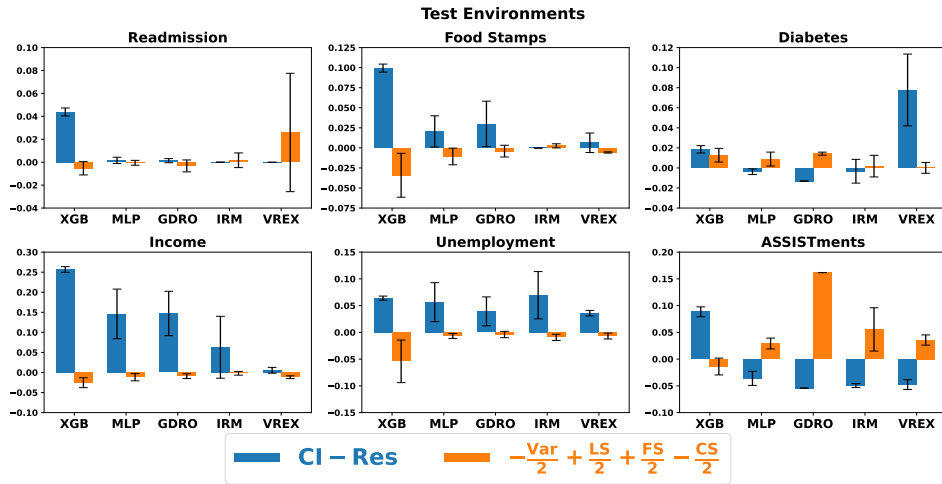


Figure C3: Dataset-specific decomposition on OOD (test-domain) data, per method.

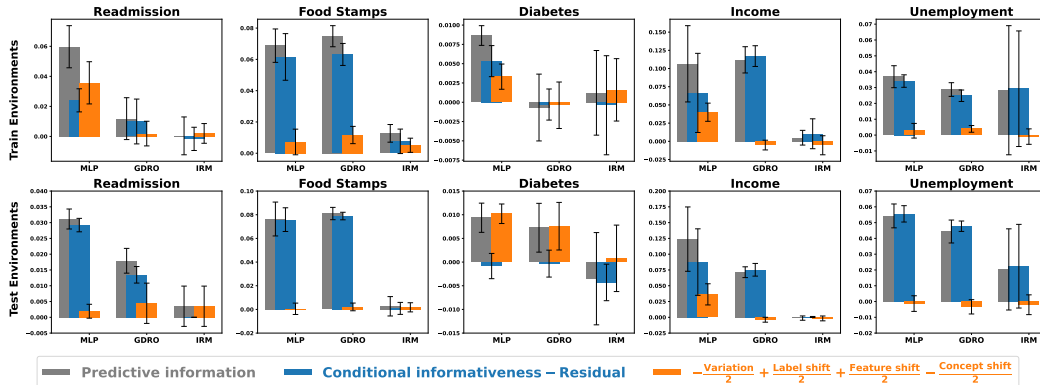


Figure C5: Dataset-specific comparison of the predictive-information terms under explicit hyperparameter tuning, for each method.

Table C2: Dataset-specific ID and OOD test accuracies (mean $\pm$ standard deviation) for each model under explicit hyperparameter tuning.

Method	Readmission		Food stamps		Diabetes	
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.
MLP	65.78 $\pm$ 0.14	61.75 $\pm$ 0.22	84.74 $\pm$ 0.03	82.00 $\pm$ 0.06	87.66 $\pm$ 0.02	83.22 $\pm$ 0.05
GDRO	60.47 $\pm$ 1.08	57.60 $\pm$ 0.42	84.47 $\pm$ 0.04	81.41 $\pm$ 0.09	87.47 $\pm$ 0.10	82.90 $\pm$ 0.10
IRM	50.28 $\pm$ 7.35	50.85 $\pm$ 1.74	80.91 $\pm$ 0.00	78.01 $\pm$ 0.00	42.54 $\pm$ 36.5	43.48 $\pm$ 31.9

	Income		Unemployment	
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.
MLP	82.92 $\pm$ 0.05	81.46 $\pm$ 0.01	97.27 $\pm$ 0.02	96.04 $\pm$ 0.05
GDRO	82.70 $\pm$ 0.05	80.16 $\pm$ 0.45	97.28 $\pm$ 0.01	96.10 $\pm$ 0.00
IRM	67.91 $\pm$ 0.00	60.20 $\pm$ 0.00	96.61 $\pm$ 0.00	94.84 $\pm$ 0.00

Figures C6–C15 present the full term-by-term information decomposition of (1) for every method and dataset, and they exhibit a consistent structure. The second column (variation) and third column (feature shift) are comparable in magnitude, so by Equation (1) their difference nearly cancels and contributes little to the total; the fourth column (label shift) and fifth column (concept shift) likewise cancel to near zero. Consequently, in almost every setting the difference conditional informativeness - residual is the dominant contributor to predictive information—exactly the pattern the confounded decomposition in Equation (4) predicts.

Hyperparameter tuning: We further corroborate the findings under explicit hyperparameter tuning. We perform a grid search and select the best configuration by ID validation accuracy—deliberately *not* by OOD validation accuracy, so that no test-domain information leaks into model selection—yielding 1500 experiments (5 datasets $\times$ 3 models $\times$ 100 sampled hyperparameter configurations). To report means and standard deviations for both accuracies and mutual information terms, each experiment is repeated over 5 random seeds, even though the seed effect is insignificant in most cases. The results (Figure C4) reproduce the main-paper conclusions: consistent with Proposition 4.1, the sum of the blue and orange terms is positively associated with model performance, and, under real-world hidden confounding shift, the conditional informativeness - residual difference contributes more to total predictive information than $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$. The dataset-specific results in Figure C5 and Table C2 exhibit the same trend.

Scope of the predictive-information decomposition: The predictive information $I(Y; \hat{Y})$ quantifies the statistical dependence between predictions and labels, which is not identical to accuracy. A binary classifier that systematically emits the *wrong* label (e.g. $\hat{Y} = 1$ whenever $Y = 0$) attains the maximal $I(Y; \hat{Y}) = 1$ bit through perfect anti-correlation while achieving 0% accuracy, precisely because mutual information measures the *reduction in uncertainty* rather than correctness. The individual terms carry analogous caveats: conditional informativeness $I(\phi(X); Y|E)$ can mask harmful variation $I(\phi(X); E|Y)$, and feature shift $I(\phi(X); E)$ can inflate $I(Y; \hat{Y})$ through spurious correlations. Within its intended scope—reasonably performant models—the decomposition nonetheless serves as an informative diagnostic, as our experiments illustrate: it reveals whether a model’s predictive information originates in generalizable patterns or in environment-specific artifacts, and so guides targeted improvements while preserving interpretability. We report these caveats explicitly to delimit the claims the decomposition can support.

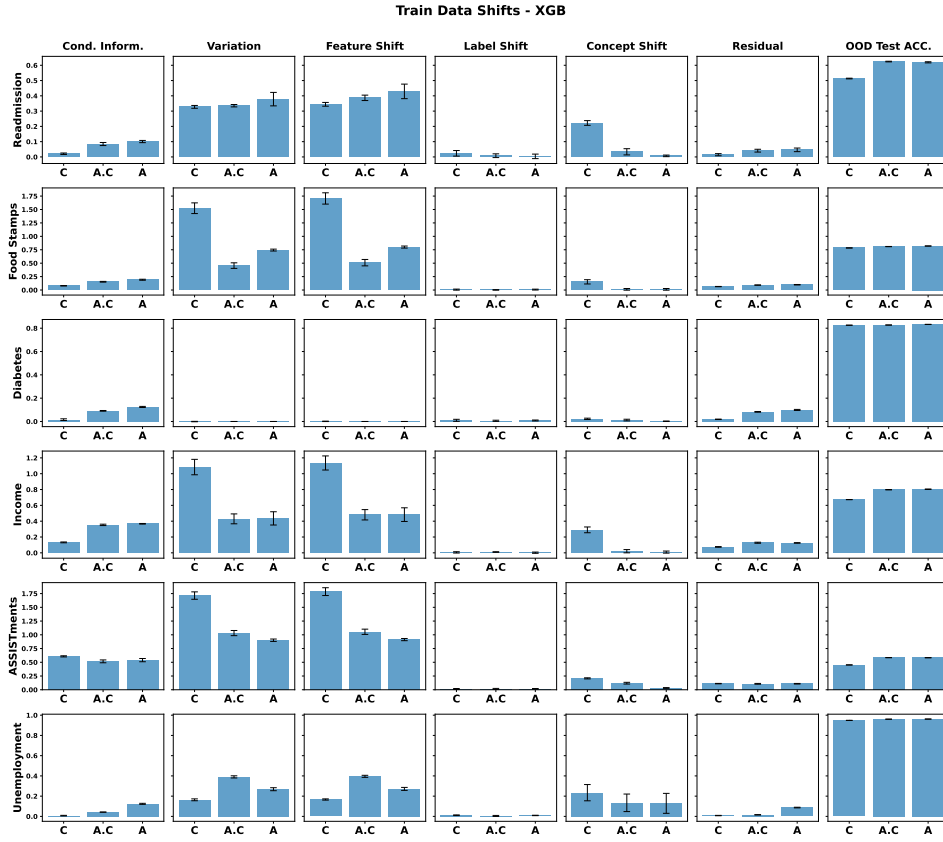


Figure C6: Term-by-term information decomposition on ID (train-domain) data for XGBoost across feature groups: (C) causal, (A.C) arguably causal, (A) all.

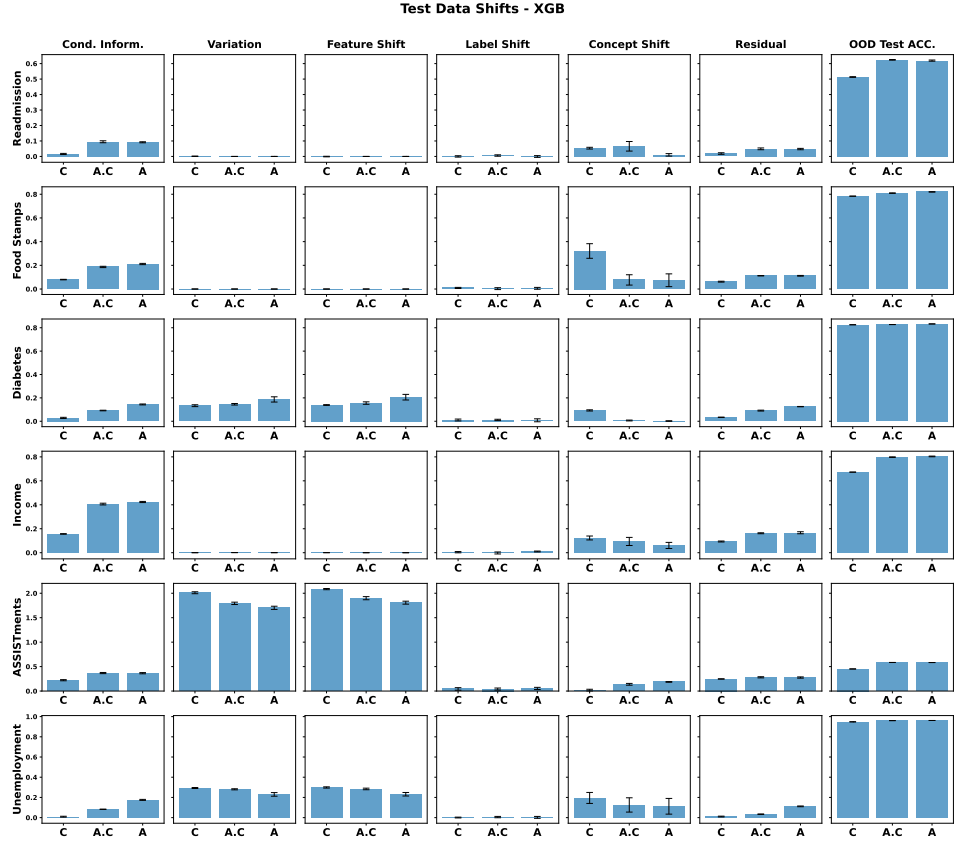


Figure C7: Term-by-term information decomposition on OOD (test-domain) data for XGBoost across feature groups: (C) causal, (A.C) arguably causal, (A) all.

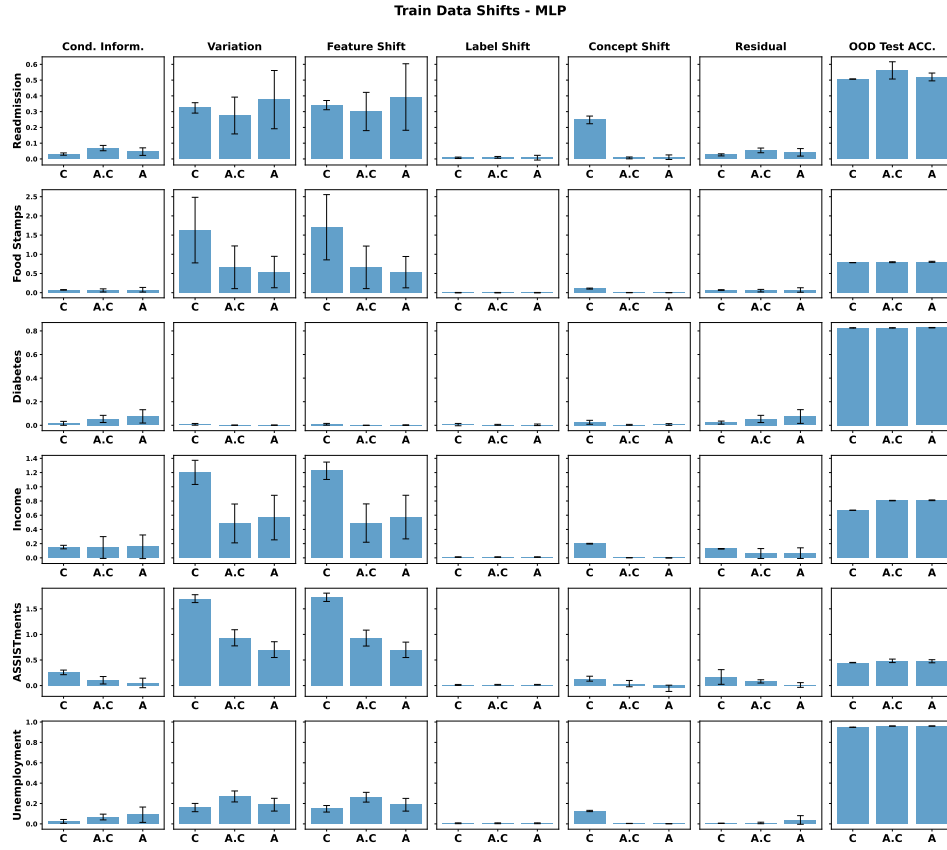


Figure C8: Term-by-term information decomposition on ID (train-domain) data for MLP across feature groups: (C) causal, (A.C) arguably causal, (A) all.

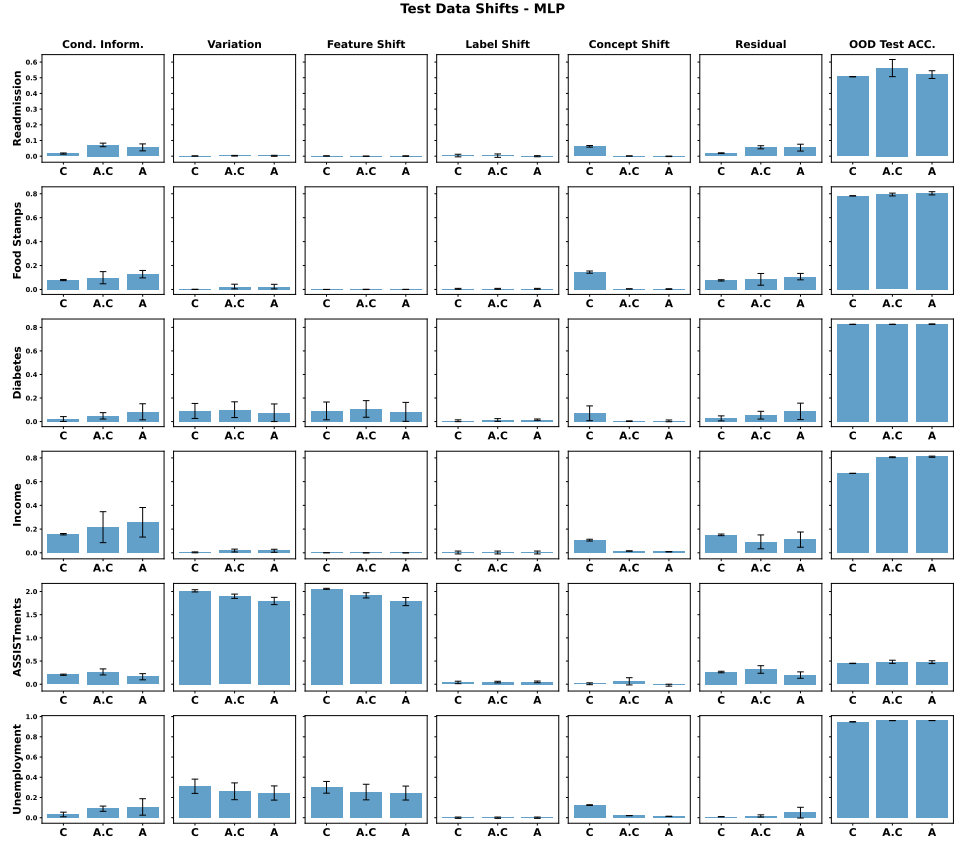


Figure C9: Term-by-term information decomposition on OOD (test-domain) data for MLP across feature groups: (C) causal, (A.C) arguably causal, (A) all.

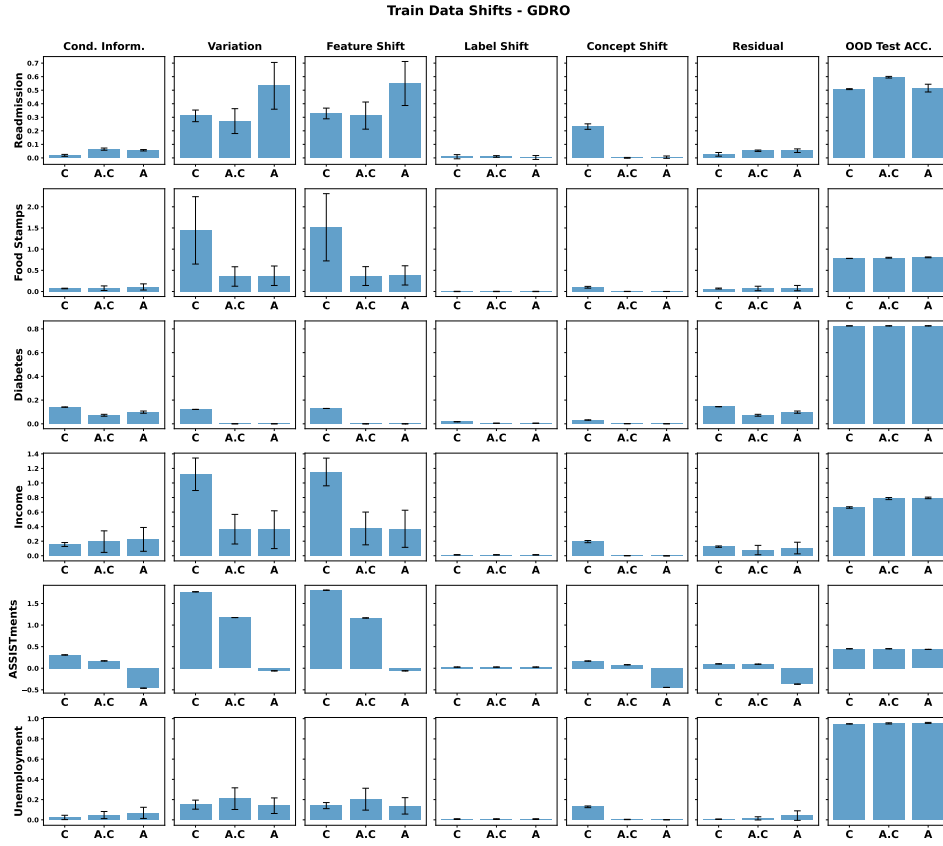


Figure C10: Term-by-term information decomposition on ID (train-domain) data for GDRO across feature groups: (C) causal, (A.C) arguably causal, (A) all.

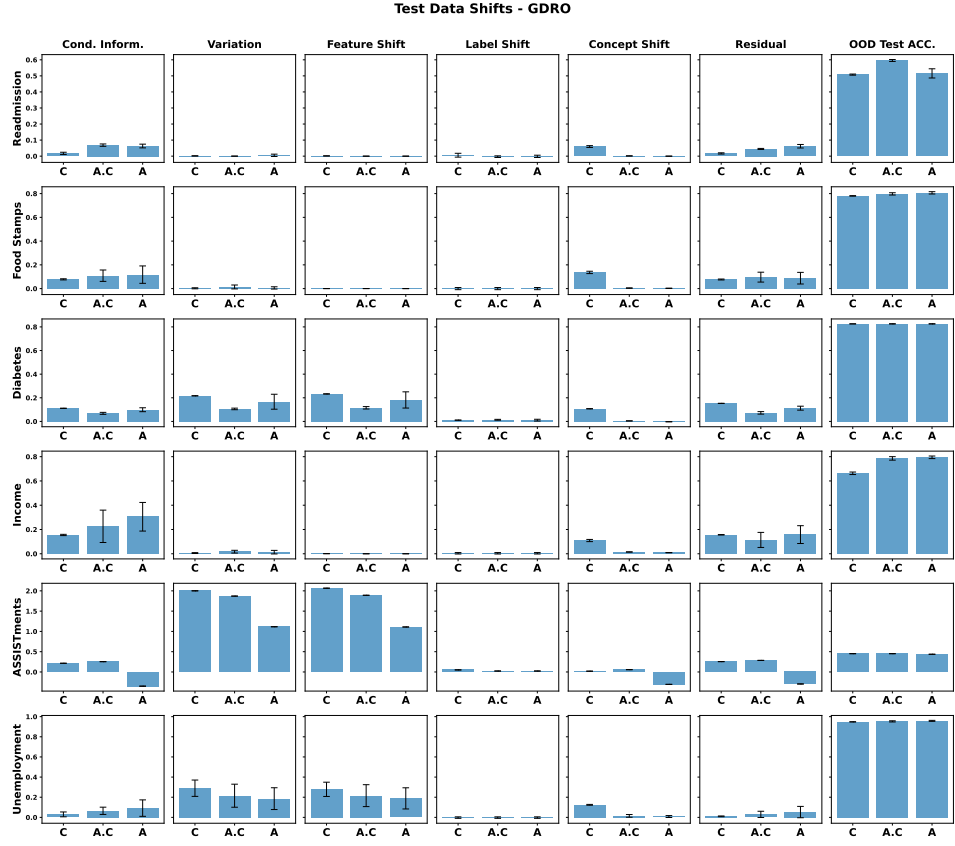


Figure C11: Term-by-term information decomposition on OOD (test-domain) data for GDRO across feature groups: (C) causal, (A.C) arguably causal, (A) all.

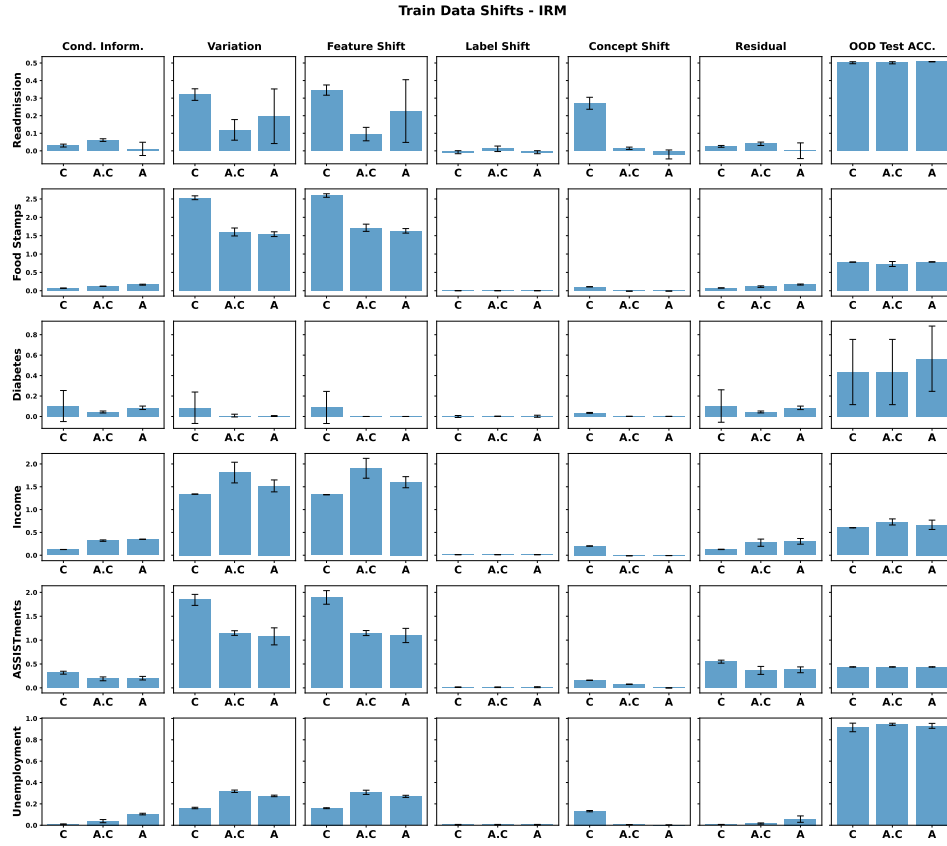


Figure C12: Term-by-term information decomposition on ID (train-domain) data for IRM across feature groups: (C) causal, (A.C) arguably causal, (A) all.

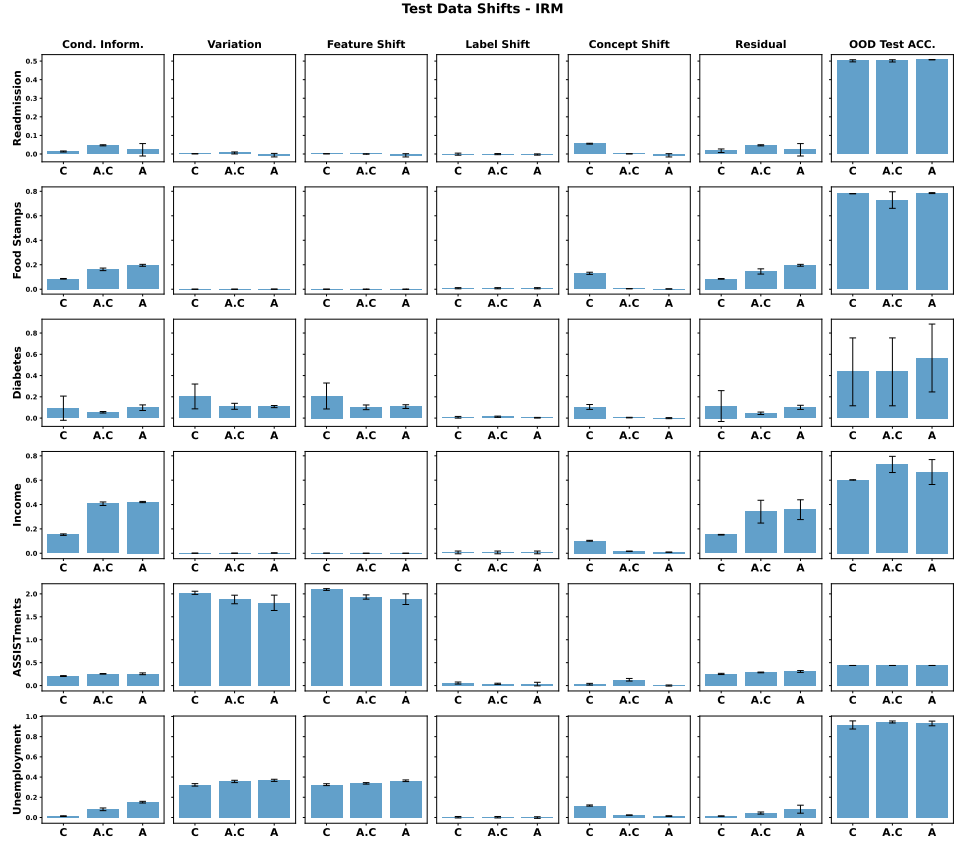


Figure C13: Term-by-term information decomposition on OOD (test-domain) data for IRM across feature groups: (C) causal, (A.C) arguably causal, (A) all.

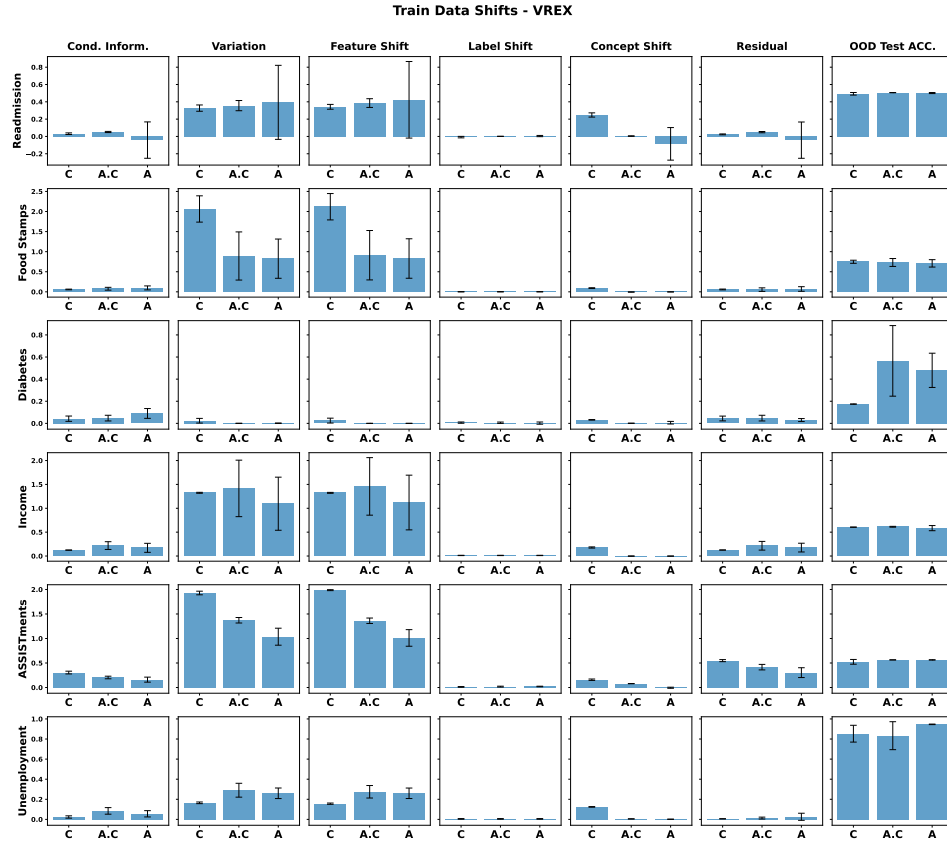


Figure C14: Term-by-term information decomposition on ID (train-domain) data for VREX across feature groups: (C) causal, (A.C) arguably causal, (A) all.

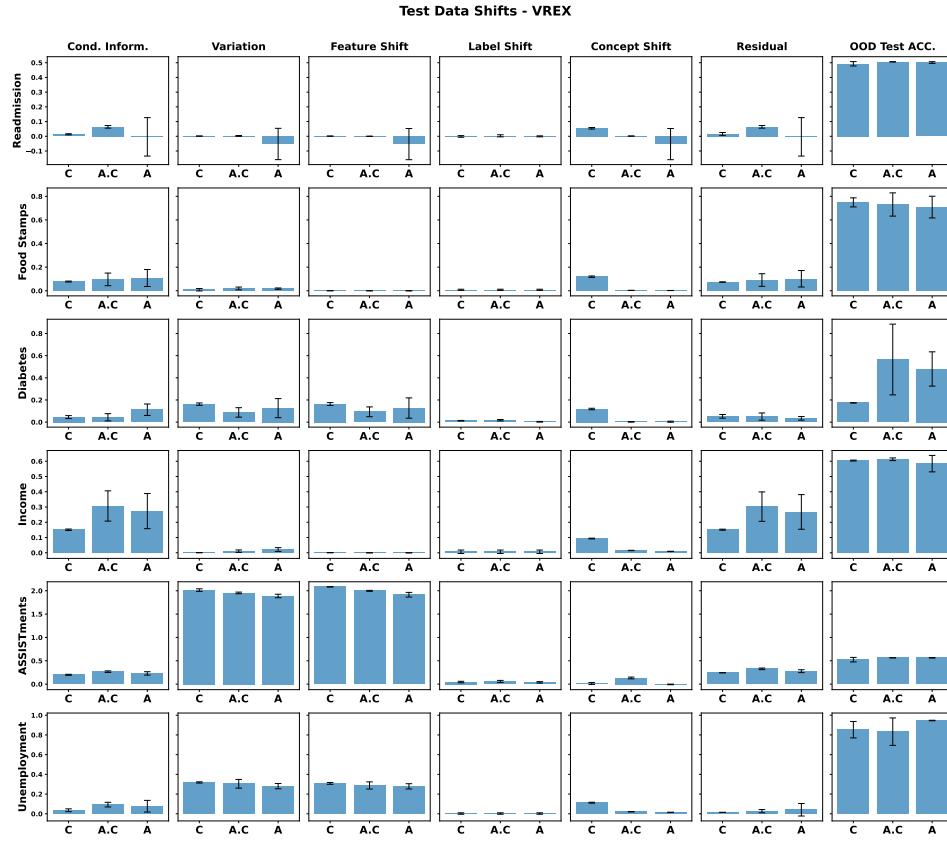


Figure C15: Term-by-term information decomposition on OOD (test-domain) data for VREX across feature groups: (C) causal, (A.C) arguably causal, (A) all.

894 D Results on synthetic datasets

895 The synthetic experiments isolate three factors under fully controlled conditions in which the con-
 896 founding structure is known by construction: (i) the effect of different informativeness criteria, (ii) the
 897 effect of the degree of confounder overlap across environments, and (iii) the effect of the ratio of
 898 causal to anti-causal variables. Because the data-generating process is specified rather than inferred,
 899 these studies furnish the interventional counterpart to the observational real-world analysis of § 5.
 900 The process follows the structural equations in (12) under the causal structure $U \rightarrow X$, $U \rightarrow Y$,
 901 $U \rightarrow \mathbf{X}_I$, and $X \rightarrow Y$, where U is the unobserved confounder, X an observed covariate, Y the
 902 target, and $\mathbf{X}_I$ the additional informative covariates.

$$\begin{aligned} U &\sim \mathcal{N}(\mu_u^e, \sigma_u) & X &\leftarrow f_X(U) + \mathcal{N}(\mu_x, \sigma_x) \\ X_i &\leftarrow f_i(U) + \mathcal{N}(\mu_i, \sigma_i); X_i \in \mathbf{X}_I & Y &\leftarrow f_Y(X, U) + \mathcal{N}(\mu_y, \sigma_y) \end{aligned} \quad (12)$$

903 We consider several functional forms for f_X , f_i , and f_Y , reporting results for each, and induce domain
 904 shift by systematically varying the environment-specific mean μ_u^e of the confounder distribution—a
 905 design that lets us dial the degree of distributional shift and observe its effect on cross-environment
 906 performance directly.

907 **Informativeness vs. accuracy:** We begin with the linear structural equations $f_X = 0.3U$, $f_i = 0.1U$,
 908 and $f_Y = X - 2U$, taking $\mu_u^e \in \{-2, 2\}$ for the training environments and $\mu_u^e \in \{0, 4\}$ for
 909 the test environments. Figure 5 of the main paper uses one hidden confounder and $|\mathbf{X}_I| = 20$
 910 informative covariates. As the number of informative covariates grows from 0 to 20, the full predicted
 911 pattern emerges: (i) decreasing mean squared error, (ii) increasing conditional informativeness,
 912 (iii) increasing feature shift, and (iv) decreasing concept shift. Two further informativeness criteria
 913 confirm the same behavior. First, with one hidden confounder and a single informative covariate
 914 ($|\mathbf{X}_I| = 1$) at noise levels $\sigma_i \in \{0, 0.1, \dots, 2.0\}$, Figure D16 shows that as the proxy noise falls from
 915 2.0 to 0.0 we again obtain (i) lower MSE, (ii) higher conditional informativeness, (iii) higher feature
 916 shift, and (iv) lower concept shift, with the variation term identically zero throughout. Second, with 20
 917 hidden confounders and 20 corresponding informative covariates under $f_X = 0.3U$, $f_i = 0.2U$, and
 918 $f_Y = X - 2U$, Figure D17 reproduces the identical trend. Across all three criteria, then, increasing
 919 informativeness consistently improves OOD generalization—exactly as Proposition 4.3 predicts.

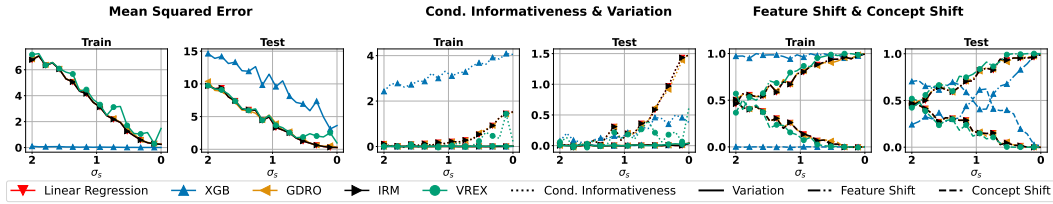


Figure D16: Adding a low-noise proxy variable reduces MSE while increasing conditional informativeness and feature shift and reducing concept shift—the lower the proxy noise, the stronger the effect.

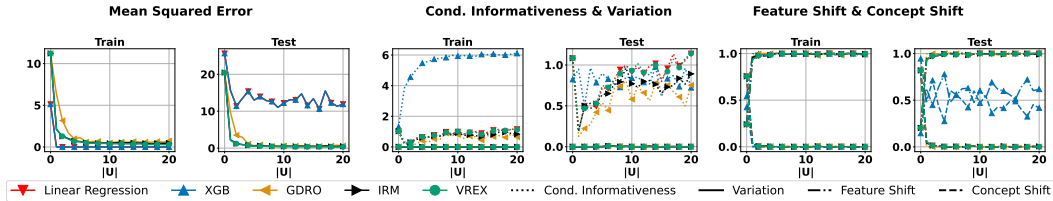


Figure D17: With a set of hidden confounders $\mathbf{U}$, adding more proxy variables $\mathbf{X}_I$ informative about Y reduces MSE while increasing conditional informativeness and feature shift and reducing concept shift.

920 **Overlapping confounder support vs. accuracy:** The method of Prashant et al. [62] assumes that
 921 the test confounder support remains within the training support. Having shown successful recovery

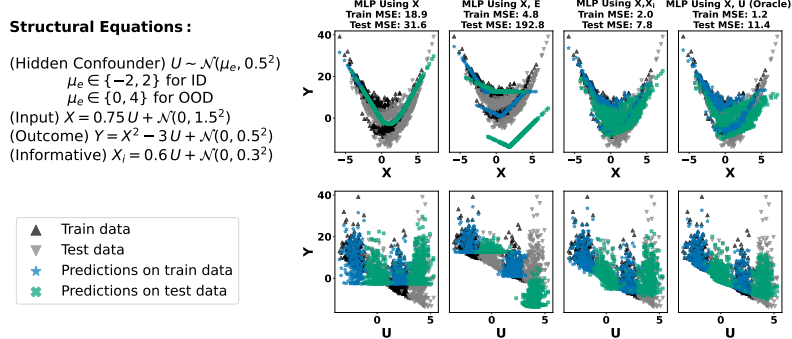


Figure D18: MLP performance in the low-overlap nonlinear setting, for the feature sets: (i) X ; (ii) X and E (environment statistics); (iii) X and X_i ; (iv) X and U (oracle).

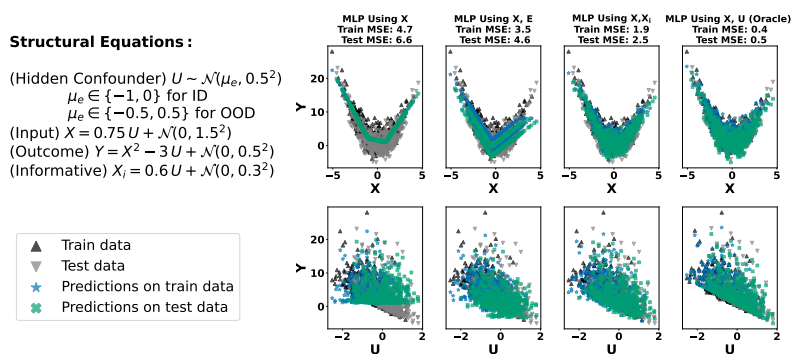


Figure D19: MLP performance in the high-overlap nonlinear setting, for the feature sets: (i) X ; (ii) X and E (environment statistics); (iii) X and X_i ; (iv) X and U (oracle).

of the correct relationships on synthetic *linear* data whenever the observed covariates carry sufficient information about the hidden confounder, we extend the analysis to *nonlinear* data with MLP and XGB models under both low- and high-overlap confounder-support conditions (Figures D18–D21). In the low-overlap regime—characterized by distant μ_e between ID and OOD data—models trained on X alone learn an inadequate global function that fails to distinguish environments. Augmenting X with environment-specific summary statistics recovers the environment-specific relationships and improves performance, and the largest gains arise once additional informative covariates are supplied, underscoring their value. For completeness, we report oracle results and U – Y scatter plots that make the learned input–output mappings explicit. Performance is, as expected, stronger under high overlap; more notably, it remains strong in the low-overlap regime—a setting that has received little prior study and that directly challenges the common-support assumption prevalent in the literature.

Number of causal vs. anti-causal variables: Recent benchmarks [24] report that baseline methods (e.g. XGB, MLP) consistently match or exceed dedicated OOD generalization techniques (e.g. GDRO, IRM, VREX) on real-world tabular data, a pattern that coincides with the prevalence of the causal structure $\mathbf{X} \rightarrow Y$. To test whether this baseline advantage is attributable to that structure, we sweep the causal-to-anti-causal ratio $\rho = |\mathbf{X}_C|/|\mathbf{X}_A| \in \{0.0, \dots, 1.0\}$, holding $|\mathbf{X}| = 50$ with $\mathbf{X} = \mathbf{X}_C \cup \mathbf{X}_A$ (so that $\rho = 0.5$ yields $|\mathbf{X}_C| = |\mathbf{X}_A| = 25$). The baselines prove resilient even under the mixed structure $\mathbf{X}_C \rightarrow Y \rightarrow \mathbf{X}_A$ (Figure D22), attaining higher PMA-OD scores [24] (the fraction of maximum OOD accuracy attained), although no single method dominates at every ρ (Figures D23, D24). To make the comparison broad, we evaluate an extensive slate of methods across three families: (1) ERM-based methods (XGBoost [15], LightGBM [34], MLP, ResNet [27], SAINT [76], TabTransformer, NODE [61], FT-Transformer, ExpGrad [2]); (2) OOD generalization methods (IRM [6], IB-IRM, IB-ERM [30], CausIRL [17], DANN [3], MMD [43], DeepCORAL [81], V-REx [38], Domain Mixup [90, 91]); and (3) robust-optimization methods (DRO [42], GroupDRO [69], Label DRO, Adversarial Label DRO [94]). These synthetic comparisons

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

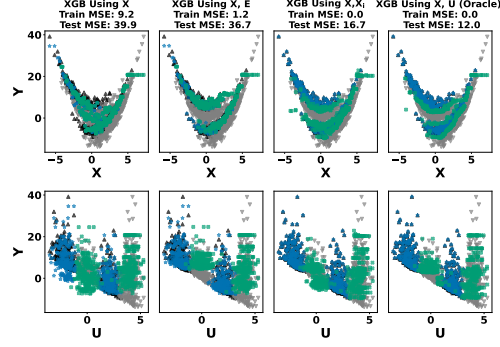


Figure D20: XGB performance in the low-overlap nonlinear setting, for the feature sets: (i) X ; (ii) X and E (environment statistics); (iii) X and X_i ; (iv) X and U (oracle).

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-1, 0\}$ for ID
 $\mu_e \in \{-0.5, 0.5\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

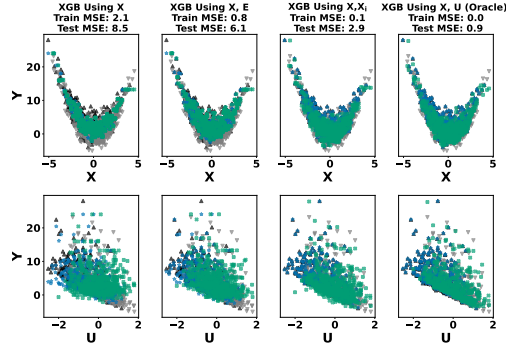


Figure D21: XGB performance in the high-overlap nonlinear setting, for the feature sets: (i) X ; (ii) X and E (environment statistics); (iii) X and X_i ; (iv) X and U (oracle).

947 are run by us under a common protocol on data we generate; the many-method real-world rankings
948 referenced above are those reported by the corresponding benchmark studies.

Percentage of Max Accuracy (PMA-OOD) on Synthetic Datasets Across All ratios ($\rho = \frac{|X_c|}{|X_s|}$)
Left: Data With Linear Equations, Right: Data With Nonlinear Equations

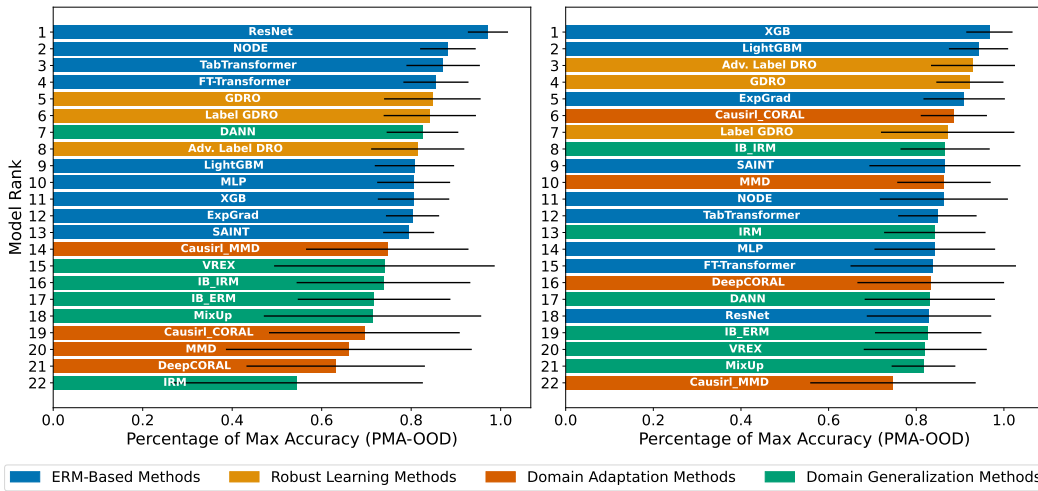


Figure D22: On the synthetic datasets, ERM-based methods attain a higher percentage of maximum OOD (PMA-OOD) test accuracy than the OOD-generalization and robust-optimization families, mirroring the real-world results of Gardner et al. [24].

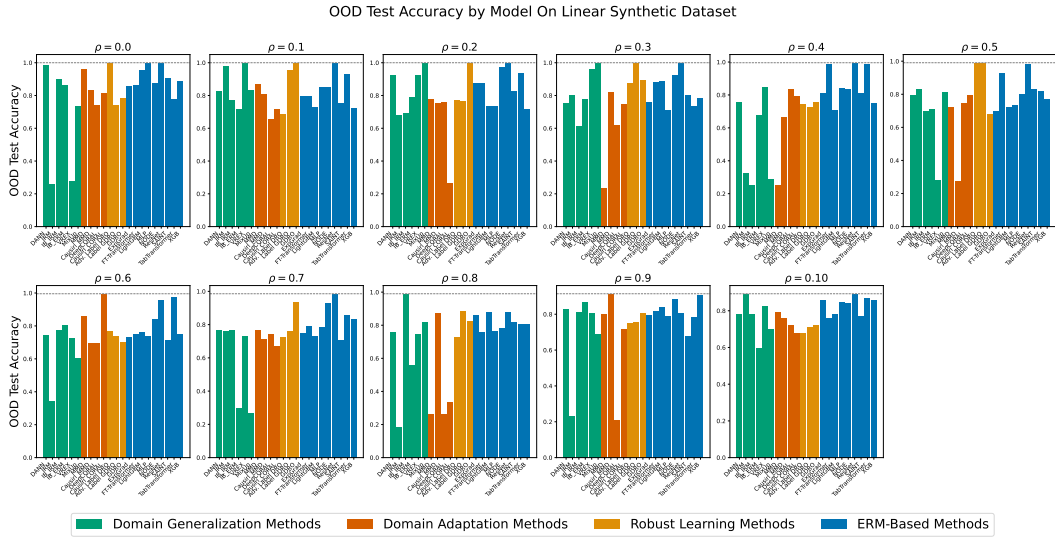


Figure D23: Per-model OOD test accuracy in the linear setting across values of the causal-to-anti-causal ratio ρ .

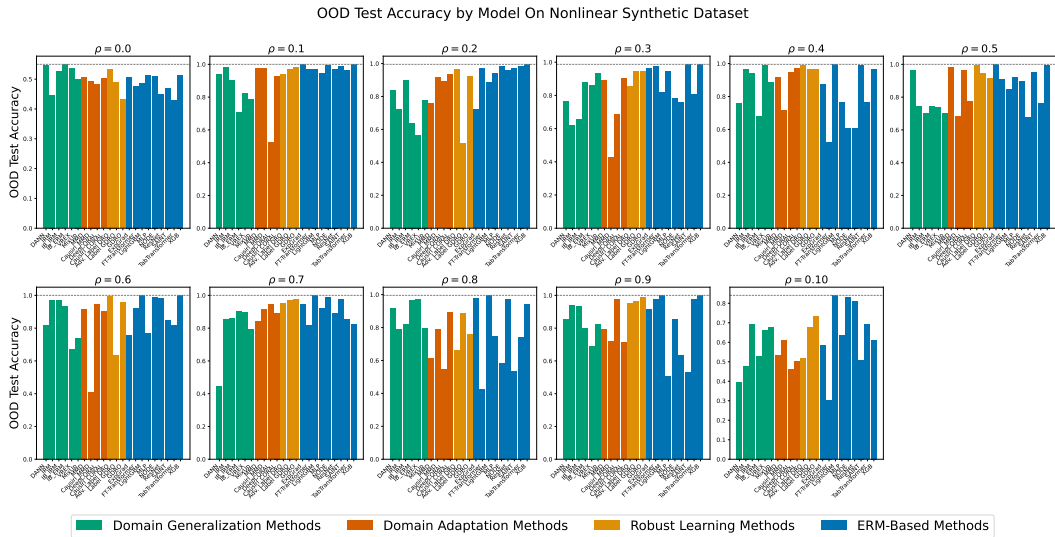


Figure D24: Per-model OOD test accuracy in the nonlinear setting across values of the causal-to-anti-causal ratio ρ .

949 E Qualitative analysis of hidden confounding shift in real-world datasets

950 This section offers a qualitative complement to the quantitative evidence of § 5, surveying *plausible*
 951 hidden confounders that could underlie the observed shifts in real-world datasets. Tables E3–E18
 952 enumerate candidate hidden confounders that might influence the observed covariates across several
 953 real-world datasets [57] (see Gardner et al. [24], Nastl and Hardt [57] for further dataset details). These
 954 candidates are generated with GPT-4o [1], and we emphasize that the resulting lists are illustrative
 955 rather than exhaustive, and are *not* independently verified. They should therefore be read as domain-
 956 informed hypotheses that make the quantitative signature in § 5 concrete and interpretable—not
 957 as evidence that any particular confounder is present. This qualitative material is deliberately kept
 958 separate from, and subordinate to, the quantitative analysis.

959 **Prompt:** We query GPT-4o for candidate hidden confounders using the prompt below.

960 For a target variable, you will be given lists of causal, arguably
 961 causal, anti-causal, and spurious covariates. Also, you will be provided
 962 with environment variables such that different datasets are collected in
 963 environments induced by these environment variables. Provide possible
 964 hidden confounders that cause any of the target, causal, arguably causal,
 965 spurious, and anti-causal covariates. That is, the hidden confounders
 966 should influence the distribution of their children in different domains.
 967 Ensure hidden confounders are not a part of the given variables, and your
 968 explanation should include how hidden confounders influence their children
 969 in different environments.

Table E3: **Dataset:** Food stamps. **Target:** Food stamp reciprocity in the past year for households with child. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Food stamp reciprocity, Age, Sex, Race, Marital status	Poor economic conditions can increase food insecurity (raising food stamp usage), while also influencing population demographics due to migration, employment patterns, and household formation.
State or regional public assistance policies	Food stamp reciprocity, Marital status, Number of children, Household income	Generous or restrictive assistance policies directly affect food stamp eligibility and indirectly influence household structure and income distribution.
Cultural attitudes towards welfare and public assistance	Food stamp reciprocity, Marital status, Disability, Race, Ethnicity	Social stigma or support for welfare affects both the uptake of food stamps and social norms around marriage, disability reporting, and identity categories.
Regional health care conditions	Food stamp reciprocity, Disability, Cognitive difficulty, Hearing difficulty, Vision difficulty	Regions with limited healthcare access may show higher disability prevalence and greater reliance on food stamps due to increased financial and care burdens.

Table E4: **Dataset:** Income. **Target:** Total person’s income $\geq 56K$ for employed adults. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Income, Occupation, Educational attainment, Marital status	Stronger local economies offer better-paying jobs and education access, increasing income; they also influence occupational choices and household formation patterns.
State or regional tax and labor policies	Income, Class of worker, Educational attainment, Marital status	Tax incentives and labor protections affect wages and employment types, while also shaping decisions about education and family due to financial security.
Cost of living and regional affordability	Income, Occupation, Marital status, Educational attainment	Higher living costs necessitate higher incomes and may drive occupational or educational shifts; they also influence marriage or cohabitation decisions.
Cultural norms and regional economic history	Income, Educational attainment, Occupation	Cultural values and historical industry presence shape education levels and job opportunities, which in turn affect income distributions.

Table E5: **Dataset:** Public coverage. **Target:** Coverage of non-Medicare eligible low-income individuals. **Environments:** Disability statuses.

Hidden confounder	Affected variables	Reason for confounding
State or regional healthcare policies	Public health coverage, Marital status, Employment status, Income	Differences in Medicaid expansion and public insurance eligibility influence health coverage; these policies also affect economic stability, employment, and family dynamics.
Economic conditions and poverty levels	Public health coverage, Employment status, Income, Marital status	Poorer regions tend to have lower employment rates and incomes, which increase reliance on public health coverage and influence marriage and household composition.
Cultural attitudes towards disability and healthcare	Public health coverage, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Stigma or support for disability and public care varies culturally, affecting both the reporting of disabilities and the likelihood of seeking or receiving public coverage.
Urban vs. rural divide	Public health coverage, Educational attainment, Marital status, Occupation	Urban areas typically offer better healthcare access, education, and jobs, all of which affect coverage likelihood and socioeconomic indicators.

Table E6: **Dataset:** Unemployment. **Target:** Classify whether a person is unemployed. **Environments:** Educational attainments.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Employment status, Occupation, Marital status, Mobility status	Regional economic strength affects job availability and unemployment rates; it also shapes occupation types, migration decisions, and household stability.
State or regional labor market policies	Employment status, Occupation, Educational attainment, Marital status	Labor regulations and unemployment benefits vary by region, affecting hiring practices, education incentives, and family structures.
Cultural and social norms around employment	Employment status, Disability status, Marital status	Cultural attitudes toward work and dependency influence unemployment reporting and societal roles around disability and family responsibilities.
Health and disability status	Employment status, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Poor health or disabilities reduce the ability to work, directly increasing unemployment and shaping associated health-related variables.

Table E7: **Dataset:** Voting. **Target:** Classify whether a person voted in the U.S. presidential election. **Environments:** United States census regions.

Hidden confounder	Affected variables	Reason for confounding
Political, family, and peer influences	Voted in national election, Party identification, Political participation	Social networks shape political ideology and engagement, influencing both voting likelihood and party alignment.
Media consumption habits	Voted in national election, Party identification, Political knowledge, Voting behavior	Media exposure affects awareness of political issues and biases, influencing party affiliation, political knowledge, and voting participation.
Social capital	Voted in national election, Political participation, Interest in elections	Strong community ties and civic networks increase political interest and participation, leading to higher voter turnout.
Civic education and political engagement programs	Voted in national election, Interest in elections, Political knowledge	Educational programs raise political awareness and civic responsibility, influencing both knowledge levels and voting decisions.
Historical and cultural context	Voted in national election, Party identification, Interest in elections	Historical events and regional political culture affect interest in elections and party alignment, which in turn influence voting behavior.

Table E8: **Dataset:** Hypertension. **Target:** Whether a person has hypertension. **Environments:** Body Mass Index (BMI) values.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status	Income, Employment status, Smoking habits, Alcohol consumption, Physical activity, Healthcare access, Medical cost, High blood pressure diagnosis	Lower socioeconomic status reduces access to healthcare and healthy lifestyle options, leading to poor diet, limited activity, and delayed diagnosis, which jointly influence both BMI and hypertension risk.
Access to healthcare services	Healthcare access, Medical costs, Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis	Limited healthcare access results in underdiagnosis and unmanaged hypertension, while also affecting lifestyle choices that vary with BMI, confounding the relationship between BMI and hypertension.
Genetic predisposition	Age group, Race, Sex, Smoking habits, Diabetes, High blood pressure diagnosis	Genetic risk factors for hypertension may co-vary with demographic attributes and influence both hypertension prevalence and BMI distribution across subpopulations.
Psychosocial stress	Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis, Diabetes, Age group	Chronic stress alters behavior (e.g., smoking, inactivity) and physiological responses, contributing to both increased BMI and elevated blood pressure, thus confounding the BMI–hypertension link.
Environmental factors	Physical activity, Diet, Smoking habits, High blood pressure diagnosis, Diabetes, BMI category	Living environments affect access to recreational spaces, food quality, and pollution exposure, influencing both BMI and hypertension risks. These vary across BMI categories, creating confounding.
Dietary habits beyond fruits and vegetables	Alcohol consumption, Smoking habits, Physical activity, High blood pressure diagnosis, BMI category	High-sodium or processed-food diets raise both BMI and hypertension risk. Variation in such unmeasured dietary habits across BMI categories creates spurious associations with hypertension.

Table E9: **Dataset:** College Scorecard. **Target:** Predict completion rate for first-time, full-time students at four-year institutions. **Environments:** Based on Carnegie Classifications.

Hidden confounder	Affected variables	Reason for confounding
Institutional funding and re-sources	Accreditor, Control of institution, Highest degree awarded, In-state tuition, Out-of-state tuition, Cost of attendance, SAT scores	Wealthier institutions can offer better academic support, facilities, and programs, leading to higher completion rates and more selective admission profiles. This varies across Carnegie classifications.
Regional socio-economic factors	Region, Poverty rate, Unemployment rate, SAT scores	Economic conditions across regions affect affordability, student preparedness, and institutional support levels, all influencing both enrollment outcomes and graduation likelihood.
Demographic factors	HBCU flag, Federal loan recipient rate, Pell grant recipient rate, ACT scores, Undergraduate enrollment	Student demographics shape financial aid needs, academic preparation, and graduation rates. Their effect differs by institution type and selectivity under Carnegie categories.
Community support and engagement	Distance-education flag, Federal loan recipient rate, Pell grant recipient rate, SAT scores, Undergraduate enrollment	Supportive institutional communities improve retention and completion. Variation in engagement across institution types and student aid profiles induces confounding.
Admission selectivity	Admission rate, SAT (reading/math) midpoints, ACT midpoint, Undergraduate enrollment	Selective admissions correlate with better-prepared students and higher completion rates, and vary with institutional prestige and classification.
State and local policies	Region, Poverty rate, Unemployment rate, Cost of attendance	Differences in education funding and public policy affect cost structures and completion outcomes, interacting with institutional classification and regional demographics.

Table E10: **Dataset:** ASSISTments. **Target:** Predict whether a student solves a problem correctly on the first attempt in an online learning tool. **Environments:** Different schools.

Hidden confounder	Affected variables	Reason for confounding
Institutional teaching quality	Hint count, Attempt count, Skill ID, Problem type, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Variation in instructional quality and pedagogy across schools affects how effectively students engage with content, leading to differences in problem-solving strategies, response behavior, and emotional states.
Student motivation	Hint count, Attempt count, Skill ID, Problem type, First action, Milliseconds to first response, Overlap time, Average confidence	Differences in intrinsic motivation across schools influence students' willingness to persevere, seek help, or give up quickly, affecting interaction and performance.
Classroom environment	Hint count, Attempt count, Tutor mode, Position, Type, First action, Average confidence	Peer dynamics, classroom culture, and noise levels vary by school and affect how confidently and independently students solve problems.
School technology infrastructure	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Access to reliable devices and fast internet differs by school, influencing response time, tool usage, and student experience.
Teacher-student interaction	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	The level of teacher guidance and feedback shapes how much support students require during problem-solving, affecting engagement and confidence.
Previous academic performance	Hint count, Attempt count, Skill ID, Problem type, First action, Average confidence	Students' prior achievement affects how easily they solve problems, their need for assistance, and their confidence, all of which vary across schools.

Table E11: **Dataset:** ICU. **Target:** Predict whether the patient will stay in the ICU for longer than 3 days. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	SES influences access to healthcare, preventive services, and overall health status. Differences in SES across insurance types lead to variability in pre-ICU health, physiological indicators, and ICU stay duration.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Hospital infrastructure and care standards affect monitoring, intervention speed, and clinical decisions. These factors vary by insurance coverage and influence ICU outcomes and vitals.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Presence of chronic conditions (e.g., diabetes, cardiovascular disease) affects both the need for prolonged ICU care and physiological measurements. The distribution of comorbidities differs across insurance types.
Insurance-related treatment variability	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	Differences in treatment timing, intensity, and access to specialists based on insurance policies affect ICU stay duration and clinical metrics.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, etc.	Inherited traits influence predisposition to organ failure, metabolic responses, and recovery trajectories. These effects are partially mediated by ethnicity and age distributions, which vary across insurance groups.

Table E12: **Dataset:** Hospital mortality. **Target:** Classify whether an ICU patient expires in the hospital during their current visit. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Systolic blood pressure, White blood cell count, etc.	SES shapes access to timely and high-quality care, preventive services, and general health status. Patients with higher SES often have better insurance and outcomes, leading to confounding with mortality risk.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, etc.	Pre-existing conditions such as diabetes or heart disease increase mortality risk and influence physiological features. Their prevalence differs by insurance type, creating confounding.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, White blood cell count, etc.	Access to advanced treatments, trained staff, and timely interventions influences survival rates. These factors correlate with insurance coverage, confounding mortality outcomes.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Genetic predispositions affect disease susceptibility and treatment responses. Variations in genetic risk factors may correlate with demographic traits across insurance types.
Lifestyle and behavioral factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Temperature, Systolic blood pressure, White blood cell count, etc.	Behaviors such as smoking, diet, and physical activity affect long-term health and mortality risk. These behaviors vary systematically with SES and insurance coverage, influencing both target and physiological features.

Table E13: **Dataset:** Childhood lead. **Target:** Predict blood lead levels above CDC blood level reference value. **Environments:** Poverty-income ratios.

Hidden confounder	Affected variables	Reason for confounding
Environmental Exposure	Blood lead levels, Country of birth, Age, Race and Hispanic origin	Environmental exposure to lead influences blood lead levels, and this varies significantly across socio-economic groups. People in lower PIR groups are more likely to live in areas with higher lead contamination, which contributes to higher blood lead levels. Furthermore, environmental factors may affect the demographic distribution (e.g., country of birth, race).
Access to Healthcare	Blood lead levels, Age, Gender, Race and Hispanic origin, Marital status, Education	Limited access to healthcare, especially in lower PIR groups, means fewer opportunities for detection and treatment of lead poisoning. This results in higher blood lead levels, with disparities also influencing demographic variables like age, gender, and education. Additionally, healthcare access varies by insurance and socio-economic status, further confounding the relationships.
Diet and Nutrition	Blood lead levels, Age, Gender, Race and Hispanic origin	Dietary factors, such as poor nutrition in lower PIR groups, can exacerbate lead absorption. Malnutrition increases the body's susceptibility to lead poisoning, raising blood lead levels. In contrast, higher PIR groups may have better access to nutritious foods, lowering lead absorption, thus creating a confounding effect in how socio-economic status and race influence lead toxicity.
Housing Conditions	Blood lead levels, Country of birth, Race and Hispanic origin, Marital status	Older housing conditions, which are more prevalent in lower PIR groups, contribute significantly to elevated lead exposure (e.g., lead paint, poor plumbing). These living conditions directly influence blood lead levels and can also correlate with demographic factors like country of birth, race, and marital status. This introduces confounding, as socio-economic status impacts both exposure and the demographics of affected individuals.
Occupation	Blood lead levels, Age, Race and Hispanic origin, Marital status, Education	Certain occupations, which are more common among lower PIR groups, involve higher lead exposure (e.g., construction, manufacturing). Occupational lead exposure directly impacts blood lead levels and is often correlated with education, marital status, and socio-economic status. The varying prevalence of lead exposure by occupation introduces confounding, especially across different PIR groups.

Table E14: **Dataset:** Diabetes. **Target:** Predict diabetes. **Environments:** Preferred race categories.

Hidden confounder	Affected variables	Reason for confounding
Genetic predisposition	Diabetes, BMI, High blood pressure, High blood cholesterol	Genetic factors and family history contribute to both the onset of diabetes and comorbid conditions like obesity, hypertension, and high cholesterol. These genetic predispositions can make individuals more susceptible to diabetes, leading to confounding as they correlate with other health indicators.
Access to healthcare	Diabetes, Physical health, BMI, Healthcare coverage, Health checkups	Limited or unequal access to healthcare, especially in marginalized racial groups, leads to disparities in diabetes diagnosis, management, and comorbidity treatment. It also influences the frequency of health checkups and access to medications, which can confound the relationship between diabetes status and other health metrics.
Dietary habits and food availability	Diabetes, BMI, Physical health, Alcohol consumption, Fruit and vegetable intake	Dietary habits, often shaped by socioeconomic status and local food environments, influence weight, health behaviors (such as alcohol consumption), and diabetes risk. People in lower socioeconomic strata may have limited access to healthy food options, leading to higher BMI and increased diabetes risk, creating confounding effects on health outcomes.
Psychosocial stress and mental health factors	Diabetes, Mental health, Physical health, BMI, Physical activity, Doctor visits	Chronic stress and mental health issues, often higher in marginalized groups, contribute to diabetes development and complicate its management. These factors also affect physical health (e.g., weight gain due to stress) and health-seeking behaviors (e.g., fewer doctor visits), leading to confounding by influencing both diabetes risk and its associated variables.
Socioeconomic status beyond income	Diabetes, Income, Physical health, Healthcare coverage, Education level	Socioeconomic factors, such as occupation, education, and neighborhood wealth, influence access to healthcare, nutrition, and overall health behaviors. These factors can confound the relationships between diabetes and other socio-economic variables like income and education, as they shape opportunities for prevention and treatment.

Table E15: **Dataset:** Sepsis. **Target:** Predict, from a set of fine-grained ICU data, whether a patient will experience sepsis onset within the next 6 hours. **Environments:** Lengths of ICU stay.

Hidden confounder	Affected variables	Reason for confounding
Infection prevalence in ICU	SepsisLabel, Temperature (Temp), Leukocyte count (WBC), Heart rate (HR), Blood urea nitrogen (BUN)	Higher infection rates in certain ICU units can lead to a higher probability of sepsis onset (SepsisLabel). These infection rates influence biomarkers such as WBC, HR, and BUN, creating confounding because the unit's infection environment affects both the likelihood of sepsis and the observed clinical measures.
Quality of ICU Care	SepsisLabel, Fibrinogen concentration (Fibrinogen), Leukocyte count (WBC), Platelet count (Platelets)	Higher-quality care in certain ICUs may lead to earlier identification and treatment of sepsis, resulting in more accurate SepsisLabel predictions. Additionally, better care could affect biomarkers like fibrinogen, WBC, and platelets, which are critical in sepsis detection and progression, thereby confounding the relationships between these variables and the outcome.
Severity of underlying conditions	SepsisLabel, Blood urea nitrogen (BUN), Creatinine, Lactate, Calcium	Patients with severe chronic conditions (e.g., kidney disease, cardiovascular issues) are at a higher risk of sepsis and may show abnormal levels in biomarkers like BUN, creatinine, lactate, and calcium. These underlying conditions contribute to the SepsisLabel outcome and confound the relationship between the biomarkers and the likelihood of sepsis, varying across ICU units depending on patient population.
Patient's socio-economic status	Age (Age), Gender (Gender), Leukocyte count (WBC), Fibrinogen concentration (Fibrinogen), Platelet count (Platelets)	Socio-economic factors, such as access to healthcare, can influence both the likelihood of sepsis and the observed clinical biomarkers. For example, patients from lower socio-economic backgrounds may have delayed hospitalizations or inadequate care, which affects both SepsisLabel and the progression of sepsis as indicated by WBC, fibrinogen, and platelet levels.
Hospital-specific protocols and treatment guidelines	SepsisLabel, Lactate, Glucose, Creatinine	Differences in hospital protocols for sepsis treatment, such as timing of interventions and choice of sepsis bundles, can affect both the SepsisLabel and biomarkers like lactate, glucose, and creatinine. These protocols lead to variability in how sepsis is diagnosed and treated across different hospitals and ICU units, confounding the relationship between biomarkers and sepsis outcomes.

Table E16: **Dataset:** Hospital readmission. **Target:** Predict whether a diabetic patient is readmitted to the hospital within 30 days of their initial release. **Environments:** Admission sources.

Hidden confounder	Affected variables	Reason for confounding
Socio-economic status (SES)	Race, Gender, Age, Payer code, Medical specialty, Number of outpatient visits, Number of emergency visits, Number of inpatient visits, Diabetes medication prescribed	SES influences access to healthcare, patient demographics, and chronic disease rates. It can also determine the type of care received based on admission source (e.g., emergency department vs. outpatient settings). Differences in healthcare access, such as availability of medications, may impact readmission rates and associated variables like outpatient visits and prescribed medication.
Severity of illness	Primary diagnosis, Secondary diagnosis, Number of diagnoses, Discharge type, Medication changes (e.g., Insulin, Glipizide)	More severe illness increases the likelihood of readmission and influences the complexity of diagnoses and the treatments administered. The severity of illness may differ based on the admission source (e.g., emergency versus outpatient), impacting the number and type of diagnoses and treatments prescribed at discharge, affecting readmission likelihood.
Access to healthcare resources	Time in hospital, Discharge disposition, Number of procedures, Number of medications, Number of lab tests	Access to healthcare resources (e.g., time in hospital, availability of procedures and medications) influences treatment decisions and outcomes. Different admission sources may have varying levels of available resources, leading to different lengths of stay, the types of procedures performed, and overall treatment quality, which can affect the likelihood of readmission.
Patient's adherence to medication	Change in medications, Diabetes medication prescribed, Number of outpatient visits, Number of emergency visits	Adherence to prescribed medications is often influenced by socio-economic status, which can vary depending on admission source. Non-adherence may lead to medication changes, affecting diabetes control and subsequent readmission risk. The number of outpatient and emergency visits can also reflect how well a patient manages their diabetes and the likelihood of complications.
Hospital-specific protocols	Time in hospital, Number of diagnoses, Discharge disposition, Readmitted	Different hospitals and healthcare systems implement various protocols for discharge planning and readmission prevention, which can affect readmission rates. These protocols may vary by admission source, where patients admitted via the emergency department may receive different follow-up instructions and care than those admitted through other channels.

Table E17: **Dataset:** MEPS. **Target:** Measure of health care utilization. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Years of education, Employment status, Hourly wage, Paid sick leave, Paid leave to visit doctor, Family size, Insurance coverage, Healthcare utilization	SES can influence access to healthcare services, insurance coverage, and the ability to use medical services. Insurance types often correlate with SES levels, where individuals with lower SES may be more likely to be insured by government programs (e.g., Medicaid), which in turn impacts healthcare utilization patterns across different SES groups.
Healthcare access	Paid sick leave, Insurance coverage, Employer offers health insurance, Healthcare utilization	Limited access to healthcare, such as lack of insurance or paid sick leave, directly impacts healthcare utilization. The type of insurance a person holds is often tied to access to various medical services. The level of coverage and accessibility differs across insurance types, influencing healthcare behaviors such as whether a patient can afford and utilize healthcare services.
Health behaviors	Perceived health status, Asthma medications, Limitations in physical functioning, Healthcare utilization	Lifestyle factors like smoking, alcohol consumption, and physical activity can directly affect health status and healthcare needs. Health behaviors differ across groups with different insurance types, and these behaviors contribute to healthcare utilization. Insurance coverage can also be influenced by perceived health status, which varies across insured groups, affecting utilization of medical services.
Chronic health conditions	Asthma medications, Perceived health status, Limitations in physical functioning, Healthcare utilization	Chronic conditions like asthma or diabetes increase the need for healthcare services, leading to higher healthcare utilization. People with chronic conditions are more likely to be covered by Medicare or Medicaid, which influences the type of insurance they hold and impacts their healthcare utilization and associated features (e.g., medications and physical limitations).
Regional healthcare infrastructure	Region, Family size, Healthcare utilization, Paid sick leave, Paid leave to visit doctor, Employment status	The quality and availability of healthcare infrastructure vary across regions, which can impact healthcare utilization. In underserved regions, people may be more reliant on public insurance options like Medicaid, affecting healthcare access and behaviors. The regional differences in healthcare systems can lead to disparities in access to medical services, insurance coverage, and utilization of healthcare.

Table E18: **Dataset:** Poverty. **Target:** Predict household income-to-poverty ratio. **Environments:** Citizenship statuses.

Hidden confounder	Affected variables	Reason for confounding
Social capital	Income, Unemployment compensation, Disability benefits, Family size, Housing conditions, Household income-to-poverty ratio	Stronger social support networks can improve access to income, benefits, and resources like unemployment or disability compensation. Social capital can vary based on citizenship status, which in turn influences income, housing conditions, and eligibility for benefits, confounding the relationship between household income and poverty ratios.
Workplace discrimination or bias	Household income, Worker's compensation, Unemployment compensation, Pension, Family income, Disability benefits	Discrimination in the workplace can limit opportunities for higher wages, pensions, or disability benefits, especially for marginalized groups (e.g., racial minorities, non-citizens). This bias varies by citizenship status and leads to biased associations between income, benefits, and poverty levels.
Healthcare access	Household income, Health insurance premiums, Medical expenses, Disability benefits, Family size, Health conditions, Medicaid/Medicare assistance	Limited access to healthcare, particularly for non-citizens, can lead to higher medical expenses and greater reliance on public health assistance. Citizenship status directly impacts eligibility for public healthcare programs, confounding relationships between medical expenses and income, and affecting poverty ratios.
Access to education and skills development	Educational attainment, Household income, Employment status, Unemployment compensation, Income from assistance, Savings	Disparities in educational opportunities, often linked to citizenship status, influence income potential and eligibility for government assistance. These disparities, in turn, affect the household income-to-poverty ratio and employment outcomes, creating confounding relationships between education and income.
Cultural factors	Family size, Household income, Savings, Disability benefits, Healthcare utilization, Income assistance, Living arrangements	Cultural factors influence financial management, family support, and the use of social programs. These factors can vary across citizenship statuses, leading to differences in how household income and assistance are distributed, and thus confounding the relationship between income, benefits, and financial behavior.
Housing market and rent conditions	Housing ownership, Housing costs, Family size, Household income, Rent payments, Social security benefits, Medical aid	Housing market conditions, especially rent disparities, can create financial strain and impact the household income-to-poverty ratio. Citizenship status often influences eligibility for housing assistance and rent subsidies, which confounds the relationship between housing costs and income, particularly in areas with significant immigrant populations.